

PRACTICE PROBLEMS

The following information relates to questions 1-5

Antelas AG is a German emerging technology company focused on manufacturing process design. The firm has developed several valuable patents related to micromachining and metal 3D printing technology and derives its revenue through licensing royalties and consultation services. Antelas recently issued the following debt instrument:

Antelas AG Four-Year Floating-Rate Notes (The “Notes”) Prospectus Summary

Issuer:	Antelas AG
Settlement Date:	[T + 5 Business Days]
Maturity Date:	[Four Years from Settlement Date]
Principal Amount:	EUR250 million
Interest:	MRR plus 250 bps p.a.
Interest Payment:	Commencing three months from [Settlement Date] to be paid quarterly with final payment on [Maturity Date]
Seniority:	The Notes are secured and unsubordinated obligations of Antelas AG and will rank <i>pari passu</i> with all other secured and unsubordinated indebtedness.
Business Days:	Frankfurt

- “Four years” and “MRR plus 250 bps p.a.” refer, respectively, to which bond features as of the settlement date?
 - Time to maturity and coupon rate
 - Coupon frequency and current yield
 - Time to maturity and contingency provision
- Which of the following statements *best* describes the source(s) of repayment for the Antelas AG floating-rate notes?
 - Since the notes are secured (or backed) by Antelas AG’s assets, the floating-rate notes would be considered asset-backed securities.
 - The Antelas AG notes rely on the operating cash flows of the firm as their *sole* source for interest and principal payments.
 - The Antelas AG notes rely on the operating cash flows of the firm as their *primary* source for interest and principal payments.
- Antelas AG’s issuer-specific credit spread rises by 25 bps p.a. one year after the FRN issuance. Which of the following *best* describes how Antelas AG’s FRN cou-

pon would change as a result?

- A. Antelas AG's FRN coupon payment to investors will *immediately* increase by 25 bps p.a.
 - B. Antelas AG's FRN coupon payment to investors will increase by 25 bps p.a. but will not become effective until the next full debt coupon period begins.
 - C. Antelas AG's FRN coupon will remain unchanged.
4. Calculate the value of Antelas's payment to bondholders on the maturity date if the MRR is 0.25% p.a. for the final interest period.
- A. EUR1,718,750
 - B. EUR251,718,750
 - C. EUR256,875,000
5. Antelas decides to raise new secured, unsubordinated debt. Which of the following best describes the priority of payments between the issuer's new debt and its existing debt?
- A. Because the Antelas AG notes have a pari passu (or "equal footing") clause, this debt obligation will be treated the same as any other secured, unsubordinated debt from this borrower.
 - B. Because the existing debt was issued earlier, it would receive priority over the new debt in terms of the payment of interest and principal.
 - C. We do not have enough information to answer this question; the necessary information would be provided in the indenture for new secured, unsubordinated notes.
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SOLUTIONS

1. A is correct. The notes mature (the date Antelas AG will pay the final coupon and principal amount) four years after the trade settlement date, and the coupon rate is variable, determined by the MRR plus a fixed 250 bp spread.
2. C is correct. Secured corporate bonds rely on the operating cash flows of the firm as their primary source for interest and principal payments and a legal claim on assets as a secondary source of payment.
A is incorrect because asset-backed securities (ABS) are issued by special-purpose entities created to take ownership in loans or receivables, the cash flows from which are used to pay interest and principal to investors.
B is incorrect because it describes *unsecured*, not secured, corporate debt.
3. C is correct. For the Antelas AG FRN, the spread is fixed at 250 bps and the coupon will only change with changes in the MRR.
4. B is correct. On the maturity date, Antelas AG pays investors the final FRN coupon payment plus the bond's principal amount. The FRN coupon comprises MRR plus the issuer-specific spread.
 - **FRN coupon rate:** MRR plus spread, or $0.25\% + 2.50\% = 2.75\%$
 - **Annual interest expense:** Principal $\times$ FRN coupon, or $\text{€EUR}250,000,000 \times 2.75\% = \text{€EUR}6,875,000$
 - **Quarterly interest expense:** Annual interest expense/4, or $\text{€EUR}6,875,000/4 = \text{€EUR}1,718,750$
 - **Principal Amount:** $\text{€EUR}250,000,000$
 - **Total Payment** = $\text{€EUR}250,000,000 + \text{€EUR}1,718,750 = 251,718,750$
5. A is correct. Antelas AG's existing secured, unsubordinated debt has a pari passu (or "equal footing") clause, which ensures that this debt obligation will be treated the same as any other secured, unsubordinated debt from this borrower.
B is incorrect because the priority of payments is unrelated to the timing of issuance.
C is incorrect because the pari passu clause ensures equal footing for similar new debt.

PRACTICE PROBLEMS

The following information relates to questions 1-5

Digistype Logistics (DILO) is a Canadian firm that generates a large portion of revenue in Europe. To finance its planned expansion into digital logistics, DILO is seeking to issue euro-denominated bonds to institutional investors in Germany. A term sheet for the proposed bond issue is presented below.

1.25% Digistype Logistics (DILO) Seven-Year Convertible Notes (the "Notes") Term Sheet	
Issuer:	Digistype Logistics Inc.
Settlement Date:	[T + 3 Business Days]
Maturity Date:	Seven Years from Settlement Date unless the Notes are redeemed earlier in a Conversion
Principal Amount:	EUR250 million
Unit Size:	EUR1,000 per bond
Interest:	3.25% fixed coupon to be paid annually starting one Year from the Settlement Date with the final coupon paid at Maturity
Seniority:	The Notes are an unsecured obligation of DILO and rank pari passu with other unsecured debt.
Conversion Provision:	An Investor has, at its sole option, the right to convert a portion or the full sum of the Principal Amount, or any part outstanding at any time at the Conversion Price during the Conversion Period.
Current Share Price:	EUR60.00 per DILO common equity share
Conversion Price:	EUR75.00 per DILO common equity share
Conversion Period:	Any Business Day starting from the Settlement Date through the Maturity Date

Institutional investors have expressed concern about increased credit risk due to Digistype's planned expansion outside of its core business. Digistype's CFO would prefer to fund the expansion using debt, despite an expectation of lower market interest rates in the next few years. The CFO is also considering removing the conversion feature because of a concern about the possible share dilution if the bonds are converted to equity and a belief that DILO shares are significantly undervalued.

1. Which of the following statements *best* describes the *type* of bonds to be issued given the issuer's country and where the bonds will be issued and traded?
 - A. Since Digistype is planning to issue and sell euro-denominated bonds, its bonds would be considered Eurobonds.
 - B. Since Digistype is planning to issue and sell euro-denominated bonds to investors in Germany, its bonds would be considered foreign bonds.
 - C. Since Digistype is planning to issue and sell bonds to domestic investors in Germany, its bonds would be considered domestic bonds.

2. Which of the following modifications could Digistype incorporate to *best* mitigate the investors' credit concerns?
 - A. Digistype could consider deferred debt coupons and/or principal payments.
 - B. Digistype could consider adding a call provision to its new debt.
 - C. Digistype could consider accelerating principal repayment or linking debt coupon changes to financial covenants.
 3. Which of the following is the *best* contingency feature Digistype could add to mitigate its own concerns regarding lower interest rates?
 - A. Add a call provision to the proposed new debt.
 - B. Add a put provision to the proposed new debt.
 - C. Add a step-up coupon to the proposed new debt.
 4. If DILO's share price on the Frankfurt Stock Exchange were to rise to EUR75, the bond's conversion value would be:
 - A. EUR1,000.
 - B. EUR15,000.
 - C. EUR75,000.
 5. Which of the following statements *best* describes the implications of removing the conversion feature from the bond?
 - A. Removal of the conversion feature would likely result in a higher coupon.
 - B. Removal of the conversion feature would likely increase the bond price.
 - C. Removal of the conversion feature would likely reduce credit risk.
-

SOLUTIONS

- B is correct. Digistrype, a Canadian firm, is planning to issue and sell euro-denominated bonds to investors in Germany. Bonds issued by entities that are incorporated in another country are called foreign bonds.

A is incorrect because Eurobonds are cross-border bonds that are not necessarily denominated in euros.

C is incorrect because domestic bonds refer to a domestic *issuer*, not investor.
- C is correct. Principal repayment acceleration, such as full amortization, partial principal amortization, or a sinking fund arrangement, is a way to reduce credit risk. Alternatively, Digistrype could incorporate debt coupon changes linked to financial covenants and/or credit ratings in order to address investor concerns.

A is incorrect because it would delay rather than accelerate repayment.

B is incorrect because a call provision does not address credit concerns.
- A is correct. A call provision gives Digistrype the right to redeem all or part of the bond prior to maturity if interest rates decline, so new debt could be issued at a lower cost.

B is incorrect because a put provision grants investors, not issuers, the right to redeem debt prior to maturity but would not be exercised if interest rates fell, because a put provision is intended to limit investor's downside, not upside.

C is incorrect; a step-up coupon is not a contingency provision.
- A is correct. The conversion value of a bond equals its par value if the current share price and conversion price are equal.

Conversion ratio = Convertible bond par value/Conversion price.

We know that

Conversion value = Conversion ratio × Current share price.

Or, after rearranging,

Conversion ratio = Conversion value/Current share price,

which can be substituted into the lefthand side of the first equation:

$$\text{Conversion value/Current share price} = \text{Convertible bond par value/Conversion price.}$$

If Current share price = Conversion price, then Conversion value = Convertible bond par value.
- A is correct. Because the conversion provision is valuable to bondholders, the convertible bond price is higher than the price of an otherwise similar bond without the conversion provision. Similarly, the yield on a convertible bond is lower than the yield on an otherwise similar non-convertible bond. Thus, should the Digistrype team elect to remove the conversion feature, it would likely have to increase the bond's coupon rate in order to attract investors.

B is incorrect because removal of the conversion feature would decrease, not increase, the bond price.

C is incorrect because removal of the conversion feature would not change the credit risk.

PRACTICE PROBLEMS

1. Match the instrument type on the left with its typical investor on the right.

A. Commercial paper	I. Insurance companies
B. Unsecured corporate bonds	II. Hedge funds
C. Secured corporate bonds	III. Money market funds

2. An issuer with a AAA credit rating would *most likely* be a:
- A. distressed issuer.
 - B. high-yield issuer.
 - C. sovereign issuer.
3. An index tracking the Bloomberg Barclays Global Aggregate Index would *most likely* have:
- A. limited security selection.
 - B. modest turnover due to changes in market values.
 - C. comprehensive exposure to fixed-income markets.
4. A firm that increases the size of an existing bond issue with a price significantly different from par is conducting a:
- A. reopening.
 - B. private placement.
 - C. distressed debt issuance.
5. The bond issue trading at the highest bid–offer spread is *most likely* a:
- A. recently issued sovereign bond.
 - B. seasoned investment-grade corporate bond.
 - C. recently issued investment-grade corporate bond.

SOLUTIONS

1.
 - A. III
 - B. I
 - C. II
2. C is correct. Sovereign government issuers typically have the lowest credit risk (highest credit rating) in each market, due to sovereign governments' right to tax economic activity. High-yield issuers have below-investment-grade credit ratings (BBB- or below), while distressed issuers typically have high-yield ratings or are in default.
3. C is correct. Mutual funds and ETFs that track this benchmark index offer *comprehensive* exposure to many fixed-income markets with *no* active security selection and portfolio turnover limited solely to either new issuance or issues no longer meeting inclusion criteria.
4. A is correct. Increasing the size of an existing bond with a price significantly different from par is referred to as the reopening of an existing bond. A private placement refers to bonds being sold to a selected investor or group of investors, versus a sale to the public. Bonds of issuers believed to be very close to or in bankruptcy are referred to as distressed.
5. B is correct. Bonds of less frequent corporate issuers or more seasoned bonds of frequent issuers are rarely traded, leading dealers to quote bid–offer spreads of at least 10–20 basis points or more for small sizes. The most recently issued, or on-the-run, developed market sovereign bonds are typically the most liquid fixed-income securities, with primary dealers making active markets in large size with bid–offer spreads equal to a fraction of a basis point. Among corporate issuers, recently issued corporate bonds from frequent issuers of higher credit quality usually exhibit the greatest liquidity and tightest bid–offer spreads.

PRACTICE PROBLEMS

1. Which of the following is a benefit to a bank that makes short-term loans and issues asset-backed commercial paper?
 - A. The bank receives commercial paper when the ABCP is issued.
 - B. Capital costs are reduced by providing undrawn backup liquidity instead of holding the short-term loans to maturity.
 - C. The bank purchases a liquid, short-term note with interest and principal payments from a short-term loan portfolio to which it would otherwise not have direct access.
2. Ewing Corp. is a large corporation that has an existing relationship with Sycamore Bank. Ewing is seeking short-term financing from a committed line of credit; however, Sycamore will offer only an uncommitted line of credit. Which of the following *best* supports Sycamore's decision?
 - A. Sycamore will receive an upfront commitment fee on the uncommitted line of credit.
 - B. Sycamore will require less bank capital for the uncommitted line than for the committed line of credit.
 - C. Sycamore can form a syndicate to reduce the amount of committed capital needed under an uncommitted line of credit.
3. Assume that today ($t = 0$) the current US five-year Treasury note trades at a price equal to the bond's face value of USD50,000,000. The security buyer takes delivery of the US Treasury note today and pays the security seller USD50,000,000. Assume a repo term of 45 days (and 360 days in a year) and a repo rate of 0.375%. If the buyer agrees to return the five-year Treasury note 45 days from today ($t = T$) to the seller, the repurchase price is *closest* to:
 - A. USD50,015,625.
 - B. USD50,023,438.
 - C. USD50,187,500.
4. A transaction where a hedge fund enters into a repo by taking a short position in a security with a view that its price will decline relative to the price of another security held is *best* described as:
 - A. a reverse repo.
 - B. a triparty repo.
 - C. financing the ownership of a security.
5. Which of the following is *not* a reason why issuers elect to issue long-term debt as opposed to short-term debt?
 - A. More stable funding for short-term activities
 - B. Lower interest rates for longer maturities for a given issuer

- C. More stable funding for long-term needs, such as capital investments
6. Which of the following statements about high yield is correct?
- A. High-yield investors benefit from higher bond prices as issuer-specific credit spreads fall, but in the case of callable debt, these gains are capped at the original purchase price.
 - B. Issuers in the high-yield market seek to retain financial flexibility by borrowing under leveraged loans with prepayment features or issuing bonds with contingency features.
 - C. Issuers who believe that their creditworthiness will decline frequently choose to issue callable debt, because the value of this contingency feature rises as a firm's borrowing costs rise.

SOLUTIONS

1. B is correct. The bank will reduce capital costs by providing undrawn back-up liquidity instead of holding the short-term loans to maturity. A is incorrect because the bank receives cash when the ABCP is issued, which increases the bank's return. C is incorrect because investors purchase a liquid, short-term note with interest and principal payments from a short-term loan portfolio to which it would otherwise not have direct access.
2. B is correct. Committed lines require more bank capital than uncommitted lines, although commitments of less than a year (usually 364 days) minimize a bank's capital requirement. A is incorrect because Sycamore will receive an upfront commitment fee on a committed line of credit. C is incorrect because Sycamore can form a syndicate to reduce the amount of committed capital needed under a committed line of credit.
3. B is correct. USD50,023,438 is calculated as:

$$\text{USD}50,000,000 \times [1 + (0.375\% \times 45/360)] = \text{USD}50,023,438.$$

In effect, the security seller borrows USD50,000,000 on a short-term basis at a low cost, with interest (USD23,438) paid at maturity, because the loan is collateralized by the US Treasury note.

A is incorrect because a repo term of 30 days, as opposed to 45 days, is incorrectly used:

$$\text{USD}50,000,000 \times [1 + (0.375\% \times 30/360)] = \text{USD}50,015,625.$$

C is incorrect because the repo term and number of days in a year are not used:

$$\text{USD}50,000,000 \times (1 + 0.375\%) = \text{USD}50,187,500.$$
4. A is correct. When viewed from the perspective of the security buyer (cash lender), a repo transaction is referred to as a reverse repurchase agreement or reverse repo. B is incorrect because under a triparty repo, both parties agree to use a third-party agent for the transaction. C is incorrect because financing the ownership of a security is not the purpose undertaken by the hedge fund.
5. B is correct. Under normal market conditions, longer maturities are associated with both higher interest rates (yields-to-maturity on government bonds) and higher credit spreads for a given issuer. A is incorrect because long-term debt provides more stable funding for short-term activities. C is incorrect because long-term debt provides more stable funding for long-term needs, such as capital investments.
6. B is correct. Issuers in the high-yield market often seek to retain financial flexibility by borrowing under leveraged loans with prepayment features or issuing bonds with contingency features, such as callable debt. A is incorrect because high-yield investors benefit from higher bond prices as issuer-specific credit spreads fall, but in the case of callable debt, these gains are capped at the call price and not the original purchase price. C is incorrect because issuers who believe that their creditworthiness will not decline frequently choose to issue callable debt, because the value of this contingency feature rises as a firm's borrowing costs fall.

PRACTICE PROBLEMS

1. Balance sheets of government entities are unlikely to include:
 - A. FX reserves.
 - B. long-term debt.
 - C. accrual of unfunded liabilities.
2. In practice, which of the following applies as it relates to sovereign government financing?
 - A. Taxpayers smooth consumption over time, saving expected future taxes today for future payment.
 - B. Taxpayers form rational expectations that today's tax cuts will result in future tax increases and pass on tax savings to descendants.
 - C. Governments seek to minimize interest rate and rollover risks by distributing debt across maturities while issuing debt in regular, predictable intervals.
3. Which of the following is correct regarding the single-price auction process?
 - A. Bidders include only dealers and institutional investors.
 - B. Competitive bidders who bid higher than the stop yield are allocated securities.
 - C. All non-competitive bids are accepted, while competitive bids are ranked starting at the lowest yield (highest bond price).
4. On-the-run sovereign debt securities are:
 - A. used for benchmark yield analyses.
 - B. issued prior to off-the-run securities.
 - C. less liquid than off-the-run securities.
5. A non-sovereign bond issued to fund public goods and services in the non-sovereign's limited jurisdiction that is repaid from local tax cash flows is referred to as:
 - A. a GO bond.
 - B. a revenue bond.
 - C. an agency bond.
6. Government agencies issue debt that is:
 - A. at a yield-to-maturity equal to that of its sovereign guarantor.
 - B. primarily repaid by cash flows related to its underlying activities.
 - C. short term, only to meet working capital and capital investment needs.

SOLUTIONS

1. C is correct. Public sector financial accounting standards vary widely and are often prepared using cash, rather than accrual-based, principles, typically excluding such items as the depreciation of fixed public goods, such as federal highways, or the accrual of unfunded liabilities, such as government pension obligations. A and B are incorrect because balance sheets of government entities are likely to include FX reserves and long-term debt.
2. C is correct. Governments seek to minimize interest rate and rollover risks by distributing debt across maturities while issuing debt in regular, predictable intervals. A and B are incorrect because taxpayers are neither able to perfectly smooth consumption over time, nor do they form rational expectations about current versus future taxes across generations, leading to fiscal instability, creating uncertainty in future tax rates, and threatening the stability of economic growth over time.
3. C is correct. All non-competitive bids are accepted, while competitive bids are ranked starting at the lowest yield (highest bond price). A is incorrect because individuals can also bid, in addition to dealers and institutional investors. B is incorrect because competitive bidders who bid higher than the stop yield (lower price) do not purchase any securities. All securities are sold for a single price, using a stop yield.
4. A is correct. The most recently issued sovereign securities, known as on-the-run securities, are used for benchmark yield analyses. B is incorrect because off-the-run securities are issued prior to on-the-run securities. C is incorrect because on-the-run securities are not less liquid than off-the-run securities.
5. A is correct. General obligation bonds are used to fund public goods and services in the non-sovereign's limited jurisdiction and are repaid from local tax cash flows. B is incorrect because revenue bonds are issued for a specific project financing (infrastructure such as roads, bridges, or tunnels) with the source of repayment often linked to a project's revenue stream (tolls, fees, etc.). C is incorrect because government agencies are quasi-government entities that issue debt in order to fund the government-sponsored provision of specific public goods or services based on sovereign or local law, with the primary source of repayment related to the public good or service and the sovereign government backing as a secondary source of repayment.
6. B is correct. For the Airport Authority of Hong Kong, for example, the primary source of repayment is cash flows from airport operations, while its sovereign government backing is a secondary source of repayment. Similarly, the primary source of repayment for Ginnie Mae is mortgage-based guaranty fees and other cash flows, with its sovereign government backing as a secondary source of repayment. A is incorrect because while sovereign agencies typically borrow at a yield-to-maturity near that of their sovereign guarantor, they do not benefit from the full liquidity premium associated with sovereign debt. C is incorrect because agencies issue a combination of short- and long-term debt to meet specific working capital and capital investment needs.

PRACTICE PROBLEMS

1. A non-callable, fixed-coupon bond has a price of 106.0625 and a YTM of 2.8%. If the YTM were to increase instantaneously by 80 bps, the price of the bond would decrease by 11%. If the YTM were to decrease instantaneously by 80 bps, the price of the bond would increase by:
 - A. less than 11%.
 - B. exactly 11%.
 - C. more than 11%.
2. Exceptions to the maturity effect exist for bonds that have:
 - A. long maturities, make small coupon payments, and trade at a discount.
 - B. short maturities, have high coupon rates, and trade at a discount.
 - C. long maturities, have high coupon rates, and trade at a premium.
3. With a constant market discount rate, as a non-callable, fixed-coupon bond approaches maturity:
 - A. the price–yield relationship will become more convex.
 - B. a low-coupon bond trading at a discount will increase in price.
 - C. a high-coupon bond trading at a premium will increase in price.
4. A fixed-income analyst is pricing a bond using the following equation.

$$PV = \frac{3}{(1 + 0.0275)^1} + \frac{3}{(1 + 0.0275)^2} + \dots + \frac{103}{(1 + 0.0275)^{40}}$$
 The bond's price, relative to par, is *most likely*:
 - A. greater than 100.
 - B. exactly 100.
 - C. less than 100.
5. A four-year, 2.0% semiannual coupon bond that pays coupons semiannually is trading at a price of 102.581. The bond's annualized yield-to-maturity is *closest to*:
 - A. 0.67%.
 - B. 1.34%.
 - C. 2.22%.
6. A bond is traded in between its coupon payment dates. Which of the following is true?
 - A. The buyer must pay the full price plus the accrued interest.
 - B. The bond's quoted price is greater than its flat price.
 - C. Accrued interest is not included in the flat price.

7. An analyst is analyzing a bond that will soon be issued by XYZ Company. The new bond will have five years to maturity and a 2.25% coupon, paid semiannually. XYZ has never issued bonds before, so the analyst will use matrix pricing. The analyst has identified four similar bonds with the following characteristics.

Bond	Maturity	Coupon (S/A)	YTM
A	4 Years	2.250%	1.6995%
B	4 Years	1.750%	1.5438%
C	7 Years	2.250%	2.2671%
D	7 Years	1.500%	1.9576%

Using the information provided, create a pricing matrix for the analyst to estimate the price of the XYZ five-year bond by filling in the matrix below with Bonds A, B, C, and D:

	Coupon Rate			
	1.50%	1.75%	2.25%	2.25%
4 Years				
5 Years				
6 Years				
7 Years				

SOLUTIONS

- C is correct. The absolute amount of the price increase would be more than the absolute amount of the price decrease for the same absolute change in YTM because of the convex relationship between bond price and yield.

A is incorrect because of the positive convex relationship between bond price and yield. For a given level of yield change, the price rise for that given amount of yield decrease will be larger than the price decline for that amount of yield increase.

B is incorrect because the relationship between bond price and yield is not linear.
- A is correct. Exceptions to the maturity effect are rare and occur only for low-coupon (but not zero-coupon) long-term bonds trading at a discount.

B is incorrect because the maturity effect holds for bonds that have short maturities and high coupons.

C is incorrect because the maturity effect always holds on bonds priced at a premium above par value.
- B is correct. Any bond trading at a discount will increase in price because of the pull-to-par effect.

A is incorrect because the price–yield relationship generally becomes less convex as a bond approaches maturity.

C is incorrect because a high-coupon bond trading at a premium will decrease in price as it approaches maturity.
- A is correct. The bond's periodic coupon rate exceeds the periodic market discount rate, or its coupon rate is "excessive" relative to the market. Therefore, the bond will trade at a premium to par.

B is incorrect. For the bond to trade at par, the coupon rate and market discount rate (yield-to-maturity) must be equal.

C is incorrect. For the bond to trade at a discount, the yield-to-maturity would have to be higher than the coupon rate.
- B is correct. Two approaches to calculating the yield-to-maturity are calculating an IRR or using the YIELD function in Microsoft Excel or Google Sheets. The periodic IRR is annualized by multiplying by two.

Period	Cash Flow	YIELD Function	
0	–102.581	Settlement date	1/1/2000
1	1	Maturity date	1/1/2004
2	1	Coupon	0.02
3	1	Price	102.581
4	1	Redemption	100
5	1	Coupon frequency	2
6	1	Yield	1.34%
7	1		
8	101		
Periodic IRR	0.67%		
Annualized IRR	1.34%		

A is incorrect because 0.67% is the periodic, not annualized, yield-to-maturity.

C is incorrect. Note that this choice can be eliminated because the bond is trading at a premium; thus its yield-to-maturity is lower than its coupon rate of 2.0%. Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

6. C is correct. The flat price, quoted by bond dealers, is the full price less accrued interest. Accrued interest is not included in the flat price.
 A is incorrect. The buyer must pay the full price, which already includes accrued interest.
 B is incorrect. The flat price is the quoted price.
7. To price the new XYZ bond, the analyst should group yields for comparable bonds in a matrix with maturities and coupon rates.

	Coupon Rate			
	1.50%	1.75%	2.25%	2.25%
4 Years		B		A
5 Years				
6 Years				
7 Years	D		C	

PRACTICE PROBLEMS

- A bond issued by RTR pays a quarterly coupon of 3.25%, has three years to maturity, and is currently trading at 97.28. The semiannual bond basis yield is closest to:
 - 1.055%.
 - 2.121%.
 - 4.242%.
- An analyst is analyzing a three-year, 2.25% annual coupon bond issued by QWE Company. Currently, the bond's yield-to-maturity is 2.707%. The three-year swap rate is 1.840%. The government spot rates are presented in the table.

Maturity	Government Spot Rate
1Y	0.899%
2Y	1.260%
3Y	1.904%

The G-Spread (in basis points) for the QWE bond is closest to:

- 80.
 - 87.
 - 135.
- The I-spread (in basis points) for the QWE bond is closest to:
 - 46.
 - 80.
 - 87.
 - If the price of the QWE bond is 98.70% of par, its Z-spread (in basis points) is closest to:
 - 80.
 - 82.
 - 87.
 - An analyst is analyzing a six-year callable bond with a semiannual coupon of 2.50% trading at a price of 98.90 per 100 par value and the following call schedule:

Call Schedule	
Period	Call Price
4–5 Years	101.50
5–6 Years	101.00

Match the yields below to the respective yield measures.

Yield	Yield Measure
2.6998%	Yield-to-first call
2.6998%	Yield-to-second call
2.9253%	Yield-to-maturity
3.1497%	Yield-to-worst

SOLUTIONS

1. C is correct. As a first step, we need to calculate the quarterly yield and annualize it:

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT+FV}{(1+r)^N}.$$

$$97.28 = \frac{0.8125}{(1+r)^1} + \frac{0.8125}{(1+r)^2} + \dots + \frac{100.8125}{(1+r)^{12}}.$$

$$r = 0.01055 \times 4 = 0.0422.$$

The next step is to convert the annualized quarterly yield to the semiannual bond basis yield.

$$\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n.$$

$$\left(1 + \frac{0.0422}{4}\right)^4 = \left(1 + \frac{APR_2}{2}\right)^2.$$

$$APR_2 = 0.04242 = 4.242\%.$$

A is incorrect because 1.0550% is the yield per quarter.

B is incorrect because 2.1100% is the yield per quarter (1.0550%) multiplied by 2.

2. A is correct. The G-spread is a yield spread above that of a government bond with the same maturity date. The yield-to-maturity for the corporate bond is 2.7070%. The yield-to-maturity for the government benchmark bond is 1.9036%.

$$\text{G-spread} = 2.707\% - 1.904\% = 0.8034\% = 80.3 \text{ bps}.$$

B is incorrect because 87 bps is the I-spread calculated as the yield spread of a bond over the standard swap rate in the same currency and with the same tenor.

C is incorrect because 135 bps is the spread calculated as the difference between the yield-to-maturity of the QWE bond and the average of government rates for all maturities.

3. C is correct. The I-spread is the yield spread of a bond over the standard swap rate in the same currency and with the same tenor. The yield-to-maturity for the corporate bond is 2.707%, and the swap rate for the same maturity is 1.840%.

$$\text{I-spread} = 2.707\% - 1.840\% = 0.867\% = 87 \text{ bps}.$$

A is incorrect because 46 bps is the difference between the bond's yield-to-maturity and coupon rate.

$$2.707\% - 2.250\% = 0.457\% = 46 \text{ bps}.$$

B is incorrect because 80 bps is the G-spread—the yield-to-maturity for the corporate bond less the yield-to-maturity for the government benchmark bond.

$$\text{G-spread} = 2.707\% - 1.904\% = 80 \text{ bps}.$$

4. B is correct. To calculate the Z-spread, we must solve for Z in the following equation, given the spot rates and price of the bond:

$$PV = \frac{PMT}{(1+z_1+Z)^1} + \frac{PMT}{(1+z_2+Z)^2} + \dots + \frac{PMT+FV}{(1+z_N+Z)^N}.$$

$$98.70 = \frac{2.25}{(1 + 0.00899 + Z)^1} + \frac{2.25}{(1 + 0.01260 + Z)^2} + \frac{102.25}{(1 + 0.01904 + Z)^3}.$$

The Solver add-in for Microsoft Excel finds $Z = 0.0082$, or 82 bps, by setting the price (sum of present values of cash flows) equal to 98.70 as the objective and Z as the change variable. Please refer to the candidate learning ecosystem online for a spreadsheet demonstrating the calculation in the downloadable Excel workbook.

A is incorrect because 80 bps is the value of the G-spread, not the Z-spread. The G-spread is calculated as the difference between the QWE bond yield and the yield of the government bond with the same maturity:

$$\text{G-spread} = 2.707\% - 1.904\% = 80 \text{ bps.}$$

C is incorrect because 87 bps is the I-spread, not the Z-spread. The I-spread is calculated as a yield spread of a bond over the standard swap rate in the same currency and with the same tenor. The yield-to-maturity for the corporate bond is 2.707%, and the swap rate for the same maturity is 1.840%.

$$\text{I-spread} = 2.707\% - 1.840\% = 87 \text{ bps.}$$

5. The correct yield values matched with the corresponding yield measures are presented in the following table:

Yield	Yield Measure
3.1497%	Yield-to-first call
2.9253%	Yield-to-second call
2.6998%	Yield-to-maturity
2.6998%	Yield-to-worst

Calculation of yield-to-first call:

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT + \text{CallPrice}}{(1+r)^N}.$$

$$PV = \frac{1.25}{(1+r)^1} + \frac{1.25}{(1+r)^2} + \dots + \frac{1.25}{(1+r)^7} + \frac{1.25 + 101.50}{(1+r)^8}.$$

$$r = 0.01548 \times 2 = 0.031497.$$

Calculation of yield-to-second call:

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT + \text{CallPrice}}{(1+r)^N}.$$

$$PV = \frac{1.25}{(1+r)^1} + \frac{1.25}{(1+r)^2} + \dots + \frac{1.25}{(1+r)^9} + \frac{1.25 + 101.0}{(1+r)^{10}}.$$

$$r = 0.014626 \times 2 = 0.029253.$$

Calculation of yield-to-maturity:

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT + FV}{(1+r)^N}.$$

$$PV = \frac{1.25}{(1+r)^1} + \frac{1.25}{(1+r)^2} + \dots + \frac{1.25}{(1+r)^{11}} + \frac{1.25 + 100.0}{(1+r)^{12}}.$$

$$r = 0.013499 \times 2 = 0.026998.$$

PRACTICE PROBLEMS

1. A two-year floating-rate note issued by a French corporation pays the three-month MRR of -0.55% plus 160 bps. The floater is priced at 101.20 per 100 of par value. Assuming the 30/360 day-count convention and 90 days per period, the discount margin for the floater is closest to:
 - A. 25 bps.
 - B. 50 bps.
 - C. 110 bps.
2. Changes in the required margin for a floater usually come from:
 - A. shifts in the yield curve.
 - B. changes in credit risk.
 - C. increases or decrease in inflation.

The following information relates to questions 3-4

A fixed-income portfolio manager is looking to value a one-year US dollar-denominated floating-rate note that has quarterly payments based on 90-day MRR plus 80 bps. Assume the following information:

90-Day MRR: 2.5%

Quoted Margin: 80 bps

Discount Margin: 100 bps

Face Value: USD100

3. Without doing any calculation, this floating-rate note is priced at:
 - A. a premium.
 - B. a discount.
 - C. par.
 4. The price of the FRN is closest to:
 - A. 98.804.
 - B. 99.804.
 - C. 100.804.
-
5. The bond equivalent yield of a 90-day European bank certificate of deposit quoted at a discount rate of 0.55% for a 360-day year is closest to:
 - A. 0.21%.

- B. 0.24%.
 - C. 0.56%.
6. A 91-day UK gilt (Treasury bill) has a face value of GBP2 million, a discount rate of 0.25%, and a price of GBP1.9 million. If a year has 366 days, the discount rate for the UK gilt is closest to:
- A. 0.1821.
 - B. 0.2011.
 - C. 0.2134.
7. Explain why a money market discount rate greater than zero understates the rate of return to an investor and understates the cost of borrowed funds for the issuer.

SOLUTIONS

1. B is correct.

Using the information provided, the periodic PMT is equal to

$$\frac{(MRR + QM) \times FV}{m} = \frac{(-0.055 + 0.016) \times 100}{4} = 0.2625.$$

Using the RATE function in Microsoft Excel, Google Sheets, or a financial calculator and using $PV = -101.20$, $FV = 100$, $PMT = 0.275$, and $N = 8$, we solve for the discount rate per period of 12.4161%. The discount margin can then be estimated by solving for DM :

$$r = \frac{MRR + DM}{m}.$$

$$0.124161 = \frac{-0.005 + DM}{4}.$$

$$DM = 0.4525.$$

The estimated discount margin is 45.25 bps.

2. B is correct. The required margin is the yield spread over or under a reference rate, reflecting the credit risk of an issuer. Changes in the required margin typically come from a change in the issuer's credit risk.
- A and C are incorrect. Both risks, as "top-down" or macroeconomic risks, would be reflected in changes in the MRR, not in a spread such as the required margin.
3. B is correct. This FRN is priced at a discount, because the quoted margin is less than the discount (required) margin.
4. B is correct. The FRN is priced at 99.804 per 100 of par value. Using the information provided, the periodic PMT is equal to

$$\frac{(MRR + QM) \times FV}{m} = \frac{(0.025 + 0.008) \times 100}{4} = 0.825.$$

The assumed discount rate per period is equal to 0.875%.

$$r = \frac{MRR + DM}{m}.$$

$$r = \frac{0.025 + 0.01}{4}.$$

$$r = 0.00875.$$

To value this FRN, take the present value of each cash flow using the discount rate per period. The FRN is priced at 99.804 per 100 of par value.

$$PV = \frac{0.825}{(1 + 0.00875)^1} + \frac{0.825}{(1 + 0.00875)^2} + \frac{0.825}{(1 + 0.00875)^3} + \frac{100.825}{(1 + 0.00875)^4}.$$

$$PV = 99.804.$$

5. C is correct. The bond equivalent yield is closest to 0.56%.

The present value of the banker's certificate of deposit is calculated as follows:

$$PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right).$$

$$PV = 100 \times \left(1 - \frac{90}{360} \times 0.0055\right).$$

$$PV = 99.865.$$

The bond equivalent yield (AOR using a 365-day year) is calculated to be approximately 0.56%:

$$AOR = \frac{\text{Year}}{\text{Days}} \times \frac{FV - PV}{PV}.$$

$$AOR = \frac{365}{90} \times \frac{100 - 99.8625}{99.8625}.$$

$$AOR = 0.0056.$$

6. B is correct. The discount rate is closest to 0.2011.

$$DR = \frac{\text{Year}}{\text{Days}} \times \frac{(FV - PV)}{FV}.$$

$$DR = \frac{366}{91} \times \frac{(2,000,000 - 1,900,000)}{2,000,000}.$$

$$DR = 0.2011.$$

7. For a money market instrument, the discount rate is calculated as

$$DR = \frac{\text{Year}}{\text{Days}} \times \frac{(FV - PV)}{FV}.$$

The denominator for the second term is FV , not PV . In theory, an interest rate is the amount earned divided by the investment amount (PV), not the maturity value (FV), which includes the earnings. So long as DR is greater than zero, FV will be higher than PV . Therefore, by design, a money market discount rate *understates* the rate of return to the investor, and it *understates* the cost of borrowed funds to the issuer.

PRACTICE PROBLEMS

The following information relates to questions 1-9

Use the spot rates provided to answer the following questions.

Government Bond Spot Rates

Maturity	Canada	Australia	Germany	Switzerland
1-year	0.3117%	0.0272%	-0.6965%	-0.7882%
2-year	0.5680%	0.0675%	-0.7034%	-0.7133%
3-year	0.7977%	0.3003%	-0.7021%	-0.6435%
4-year	0.9640%	0.5853%	-0.6418%	-0.5615%
5-year	1.1105%	0.8052%	-0.5578%	-0.4757%
6-year	1.2052%	0.9877%	-0.4796%	-0.4080%
7-year	1.2998%	1.1702%	-0.4014%	-0.3402%
8-year	1.3935%	1.2878%	-0.3269%	-0.2673%
9-year	1.4872%	1.4054%	-0.2524%	-0.1945%
10-year	1.5809%	1.5230%	-0.1779%	-0.1216%

- Which level of coupon would the Swiss government choose if its goal is to minimize the premium over par value for a five-year debt issue?
 - 1.0%
 - 0.5%
 - 0.0%
- The Australian government is considering issuing three-year bonds with a 3.0% coupon with annual payments. Given the spot rates in Australia, the price is *closest* to:
 - 108.05.
 - 108.07.
 - 108.69.
- Suppose the one-year and two-year spot rates in Canada move upward by 20 bps but the three-year Canadian spot rate remains unchanged. Which of the following best reflects the updated three-year Canadian par rate?
 - Increases by approximately 13 bps
 - Increases by approximately 7 bps
 - Increases by approximately 0.16 bps

4. Match the spot rates needed to calculate the corresponding implied forward rates.

1. Two-year and five-year spot rates	A. Two-year implied forward rate in five years
2. Two-year and seven-year spot rates	B. Three-year implied forward rate in two years
3. Five-year and seven-year spot rates	C. Five-year implied forward rate in two years

5. Which of the following choices is the closest to the implied five-year forward rate in two years for Swiss government bonds?

- A. -0.191%
- B. -0.001%
- C. -0.317%

6. Match the sequence of forward rates and the corresponding spot rate that the forward rates will calculate.

Forward rates	Spot rate calculation
1. $0y1y \times 1y1y \times 2y1y$	A. Four-year spot rate
2. $0y1y \times 1y2y \times 3y3y$	B. Three-year spot rate
3. $0y1y \times 1y1y \times 2y1y \times 3y1y$	C. Six-year spot rate

7. Which of the following statements about yield curves is true?

- A. Spot curves are derived from par rates.
- B. Par curves are derived from forward rates.
- C. Forward curves are derived from spot rates.

8. Match the following statements about forward rate curves with corresponding statements about spot rate curves.

1. The forward rate curve lies above the spot rate curve.	A. The spot rate curve is downward sloping (i.e., inverted).
2. The forward rate curve lies below the spot rate curve.	B. The spot rate curve is upward sloping (i.e., normal).
3. The forward rate curve is equivalent to the spot rate curve.	C. The spot rate curve is flat across all maturities.

9. Match the following statements about par rate curves with corresponding state-

ments about spot rate curves.

1. The par rate curve lies above the spot rate curve.	A. The spot rate curve is upward sloping (i.e., normal), and spot rates are negative.
2. The par rate curve lies below the spot rate curve.	B. The spot rate curve is upward sloping (i.e., normal), and spot rates are positive.
3. The par rate curve is equivalent to the spot rate curve.	C. The spot rate curve is flat across all maturities.

SOLUTIONS

1. C is correct. Given that Switzerland's current five-year spot (and par) rate is negative, any non-negative coupon will cause the Swiss government to issue debt at a premium to par. Issuance of debt without a coupon will minimize the size of the premium.

2. B is correct.

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \dots + \frac{PMT + FV}{(1+Z_N)^N}.$$

$$PV = \frac{3}{(1+0.000272)^1} + \frac{3}{(1+0.000675)^2} + \frac{103}{(1+0.003003)^3}.$$

$$PV = 108.07.$$

3. C is correct. In Example 2, we calculated the three-year Canadian par rate. The par rate of 0.7952% approximates the spot rate of 0.7977%, so the most important change affecting the three-year par rate is the change in the three-year spot rate. Because it does not change, the only change will be a slight move in the par rate toward the spot rate (to approximately 0.7968%).

4. 1. B. By using two-year and five-year rates, the difference represents the tenor of the implied forward rate of three years, and the shorter tenor rate reflects the beginning of the forward in two years.
2. C. By using two-year and seven-year rates, the difference represents the tenor of the implied forward rate of five years, and the shorter tenor rate reflects the beginning of the forward in two years.
3. A. By using five-year and seven-year rates, the difference represents the tenor of the implied forward rate of two years, and the shorter tenor rate reflects the beginning of the forward in five years.

5. The correct answer is A. The implied five-year forward rate in two years is the 2y5y rate. To calculate, we use the seven-year Swiss government rate and the two-year Swiss government rate, as follows:

$$2y5y \text{ IFR} = [(1 - 0.003402)^7 / (1 - 0.007133)^2]^{(1/5)} - 1 = 0.191\%.$$

Choice B represents the 5y2y implied forward rate, and Choice C represents the 2y3y implied forward rate.

6. 1. B. This is a series of three one-year forwards, thus providing the information for a three-year spot rate.
2. C. This sequence is a one-year spot rate, followed by a two-year forward rate in one year, followed by a three-year forward rate in three years.
3. A. This sequence is a series of four one-year forwards.
7. The correct answer is C. The forward rates used to construct forward curves are derived from spot rates. Statement A is not true, because par rates are derived from spot rates. Statement B is not true, because par curves are derived from spot rates.
8. 1. B. In a normal interest rate environment with an upward-sloping yield curve, forward interest rates will be greater than their corresponding spot rates across

all maturities.

2. A. In an inverted spot rate environment, forward interest rates will be below their corresponding spot rates.
 3. C. A flat maturity structure of interest rates is sometimes assumed. In such an environment in which spot rates are constant, this implies that future expectations are for interest rates to remain at a constant level. Thus, forward rates equal spot rates in a flat term structure environment.
- 9.
1. A. This situation is illustrated in the German and Swiss government interest rate data in which spot rates are negative while the spot rate curve is upward sloping. In such instances, par rates lie above spot rates.
 2. B. This situation is illustrated in the Canadian and Australian government interest rate data in which spot rates are positive and the spot curve is normal.
 3. C. This is a hypothetical case but can be seen by creating a flat spot rate curve.

PRACTICE PROBLEMS

1. Assuming no capital gain or loss when the bond is sold, an investor's horizon yield is equal to a bond's yield-to-maturity at the time of investment if coupons are reinvested at a rate:
 - A. lower than the bond's yield-to-maturity at investment.
 - B. equal to the bond's yield-to-maturity at investment.
 - C. higher than the bond's yield-to-maturity at investment.
2. An investor purchases an eight-year, 6.4% annual coupon eurobond priced at par (settlement: 15 June 2031; maturity: 15 December 2039) and sells it after six years. Assuming interest rates rise by 100 bps immediately after purchase, the investor's rate of return at the end of six years is:
 - A. lower than 6.4%.
 - B. equal to 6.4%.
 - C. higher than 6.4%.
3. The carrying value of a bond purchased at a price below par is equal to the original purchase price:
 - A. minus the accumulated premium amortization.
 - B. minus the accumulated discount amortization.
 - C. plus the accumulated discount amortization.
4. The duration gap at which reinvestment risk and price risk are equal is:
 - A. less than zero.
 - B. equal to zero.
 - C. greater than zero.
5. When an investor's investment horizon is less than a bond's Macaulay duration, the investor faces downside risk if, after the purchase of the bond, interest rates:
 - A. decrease.
 - B. increase.
 - C. either decrease or increase.
6. An investor purchases a 12-year, 5.8% annual bond and intends to sell it after 10 years. The Macaulay duration of the bond is 8.97 years. If interest rates fall by 75 bps immediately after the purchase of the bond, the investor faces:
 - A. negative reinvestment risk.
 - B. no reinvestment risk.
 - C. positive reinvestment risk.

7. Consider a bond that has three years remaining to maturity, a coupon of 4% paid semiannually, and a yield-to-maturity of 4.60%. Assuming it is 12 days into the first coupon period and a 30/360 basis, the bond's annualized Macaulay duration is *closest* to:
- A. 1.8764 years.
 - B. 2.8386 years.
 - C. 2.8553 years.
8. Hightest Capital purchases a seven-year, 6.4% coupon bond and has an intended investment horizon of four years. The Macaulay duration of the bond is 5.86 years. If interest rates increase by 50 bps immediately after buying the bond, Hightest Capital faces:
- A. negative price risk.
 - B. negative reinvestment risk.
 - C. positive price risk.

SOLUTIONS

1. B is correct. For the horizon yield to match the original yield-to-maturity of the bond, two conditions must be met:

(1) The coupon payments are reinvested at the original yield-to-maturity.

(2) The bond is sold at the constant-yield price trajectory, implying no capital gain or loss when the bond is sold.

Since there is no gain or loss when the bond is sold, the horizon yield will be equal to the yield-to-maturity if the first condition is met (i.e., the coupons are reinvested at the original yield-to-maturity).

A is incorrect because the horizon yield will be lower than the yield-to-maturity if the coupons are reinvested at a lower rate than the original yield-to-maturity, assuming no capital gain or loss on sale of the bond.

C is incorrect because the horizon yield will be higher than the yield-to-maturity if the coupons are reinvested at a higher rate than the original yield-to-maturity, assuming no capital gain or loss on sale of the bond.

2. A is correct. The future value of reinvested coupon interest is

$$= FV(0.074, 6, 6.4, 0, 0) = 46.245.$$

The sale price of the bond at the end of six years is

$$= PV(0.054, 2, 6.40, 100, 0) = 98.202,$$

which results in a six-year horizon yield of 6.32%, which is lower than 6.40%:

$$r = \left(\frac{FV}{PV} \right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{46.245 + 98.202}{100} \right)^{\frac{1}{6}} - 1.$$

$$r = 0.0632 = 6.32\%.$$

B and C are incorrect, because interest rates have risen and the investor's investment horizon is not long enough to offset the price decline with additional reinvestment return.

3. C is correct. A bond purchased at a discount will have its price "pulled to par" as it approaches maturity. Its carrying value at time t is equal to its purchase price plus the accumulated discount amortization.

A is incorrect because a bond purchased below par is purchased at a discount, not a premium.

B is incorrect because the accumulated discount amortization is added, not subtracted, from the purchase price to calculate carrying value.

4. B is correct. Reinvestment risk and price risk equally offset each other when an investor's investment horizon equals a bond's Macaulay duration. This is when the duration gap is zero.

A is incorrect because a duration gap of less than zero means that Macaulay duration is less than the investor's investment horizon and that reinvestment risk is higher than price risk.

C is incorrect because when the duration gap is greater than zero, the Macaulay duration is greater than the investor's investment horizon, so price risk is higher than reinvestment risk.

5. B is correct. If the investment horizon is less than the bond's Macaulay duration, then the investor's risk is higher interest rates. For such an investor, the market price risk dominates the coupon reinvestment risk.
A is incorrect because an investor faces risk due to a lower interest rate when the investment horizon is greater than the bond's Macaulay duration.
C is incorrect because when the investment horizon is less than the Macaulay duration, the investor faces risk due to higher interest rates, not lower interest rates.
6. A is correct. The investor has an investment horizon of 10 years, which is greater than the Macaulay duration of 8.97 years. Therefore, reinvestment risk dominates price risk.
B is incorrect because the investor faces reinvestment risk because the investment horizon is higher than the Macaulay duration.
C is incorrect because the investor faces negative coupon reinvestment risk, not positive coupon reinvestment risk, since the coupons will be reinvested at a lower interest rate.
7. B is correct.

Period	Time to Receipt	Cash Flow	PV	Weight	Time to Receipt × Weight
1.0000	0.9333	2.0000	1.9580	0.0199	0.0186
2.0000	1.9333	2.0000	1.9140	0.0194	0.0376
3.0000	2.9333	2.0000	1.8709	0.0190	0.0557
4.0000	3.9333	2.0000	1.8289	0.0186	0.0730
5.0000	4.9333	2.0000	1.7878	0.0182	0.0896
6.0000	5.9333	102.0000	89.1261	0.9050	5.3695
			98.4856	1.0000	5.6439
			Annualized MacDur		2.8220

8. A is correct. Hightest Capital's investment horizon is four years, which is less than the bond's Macaulay duration of 5.86 years. Therefore, price risk dominates reinvestment risk and Hightest Capital faces price risk from rising interest rates.
B is incorrect because the increase in interest rates is beneficial for coupon reinvestment. Therefore, Hightest Capital has positive reinvestment risk.
C is incorrect because Hightest Capital faces the risk that the price of the bond will fall as a result of the 50 bp increase in interest rates and therefore has negative price risk.

PRACTICE PROBLEMS

1. *True or false:* Curve duration is a type of yield duration.

A. True
B. False

The following information relates to questions 2-7

A portfolio manager is assessing the interest rate risk of three bonds as she considers making an investment of USD50 million. All three bonds are issued on 1 June 2026 and mature on 1 June 2030, and they have the following characteristics:

Characteristic	Bond One	Bond Two	Bond Three
Coupon (semiannual)	7%	3%	5%
Yield-to-maturity	3%	7%	5%

2. The Macaulay duration, in years, for Bond One is *closest* to:
- A. 3.54.
B. 3.59.
C. 3.78.
3. The modified duration for Bond Two is *closest* to:
- A. 3.59.
B. 3.65.
C. 3.78.
4. Relative to a five-year zero-coupon bond priced to yield 5%, Bond Three has a modified duration that is *best* described as:
- A. lower than the zero-coupon bond's modified duration.
B. the same as the zero-coupon bond's modified duration.
C. greater than the zero-coupon bond's modified duration.
5. If the yields-to-maturity for all three bonds were to increase by 100 bps, which bond has the greatest anticipated decrease in price?
- A. Bond One
B. Bond Two
C. Bond Three

6. Which bond has the highest price value of a basis point?
 - A. Bond One
 - B. Bond Two
 - C. Bond Three

 7. The portfolio manager is interested in comparing the interest rate risk of Bond Three to that of Bond Four, a floating-rate note that resets every six months. On 1 June 2026, both bonds were priced to yield 5%. If the yield changes from 5% to 5.25% halfway through the first coupon period, which bond has the greater Macaulay duration?
 - A. Bond Three
 - B. Bond Four
 - C. Neither: The Macaulay duration is the same for both bonds.
-

SOLUTIONS

1. B is correct. The Macaulay, modified, and money duration measures, as well as the price value of a basis point, are all types of yield duration. The curve duration is not classified as a type of yield duration.
2. B is correct.

Period	Time to receipt	Cash flow	Present value	Weight	Time to receipt × weight
1	1.0000	3.5	3.44828	0.02999	0.02999
2	2.0000	3.5	3.39732	0.02955	0.05910
3	3.0000	3.5	3.34711	0.02911	0.08734
4	4.0000	3.5	3.29764	0.02868	0.11473
5	5.0000	3.5	3.24891	0.02826	0.14129
6	6.0000	3.5	3.20090	0.02784	0.16704
7	7.0000	3.5	3.15359	0.02743	0.19200
8	8.0000	103.5	91.87810	0.79914	6.39308
			114.97185	1.00000	7.18458

Annualized Macaulay duration 3.59229

3. B is correct.

Period	Time to receipt	Cash flow	Present value	Weight	Time to receipt × weight
1	1.0000	1.5	1.44928	0.01680	0.01680
2	2.0000	1.5	1.40027	0.01623	0.03247
3	3.0000	1.5	1.35291	0.01569	0.04706
4	4.0000	1.5	1.30716	0.01516	0.06062
5	5.0000	1.5	1.26296	0.01464	0.07321
6	6.0000	1.5	1.22025	0.01415	0.08488
7	7.0000	1.5	1.17899	0.01367	0.09568
8	8.0000	101.5	77.08027	0.89366	7.14930
			86.25209	1.00000	7.56003

Annualized Macaulay duration 3.780016

Modified duration 3.652190

4. A is correct. No calculation is necessary, as we know that a five-year zero-coupon bond has a Macaulay duration of 5, which is greater than that of a coupon bond like Bond three. Since they have the same yields, the bond's modified durations will have the same relationship as their Macaulay duration.
Alternatively, we can calculate Bond Three's modified duration of 3.585:

Period	Time to receipt	Cash flow	Present value	Weight	Time to receipt × weight
1	1.0000	2.5	2.43902	0.02439	0.02439
2	2.0000	2.5	2.37954	0.02380	0.04759
3	3.0000	2.5	2.32150	0.02321	0.06964
4	4.0000	2.5	2.26488	0.02265	0.09060
5	5.0000	2.5	2.20964	0.02210	0.11048
6	6.0000	2.5	2.15574	0.02156	0.12934
7	7.0000	2.5	2.10316	0.02103	0.14722
8	8.0000	102.5	84.12652	0.84127	6.73012
			100.00000	1.00000	7.34939
			Annualized Macaulay duration		3.6747
			Modified duration		3.5851

The zero-coupon's modified duration is $5/(1 + 0.05) = 4.7619$.

5. B is correct.

	Bond One	Bond Two	Bond Three
Full price (percent of par)	114.972	86.252	100.000
Modified duration	3.539	3.652	3.585
Investment (in USD millions)	50	50	50
Money duration	176.960	182.609	179.253
Change in value (in USD millions)	−1.770	−1.826	−1.793

6. A is correct.

	Bond One	Bond Two	Bond Three
Price	114.972	86.252	100.000
Price + 1 bp	114.931	86.221	99.964
Price − 1 bp	115.013	86.284	100.036
PVBP	0.04069	0.03150	0.03585

7. A is correct. The Macaulay duration for Bond Three is 3.6747. The Macaulay duration for Bond Four is 0.5 because there is exposure to interest rate changes for one-half of the coupon period.

PRACTICE PROBLEMS

1. A 5.5% semiannual-pay fixed-coupon bond is issued at par on 1 May 2025 and matures on 1 May 2029. For a 50 bps increase and decrease in yield-to-maturity, PV_+ and PV_- are 99.82283 and 100.177546, respectively. The approximate convexity is *closest* to:
 - A. 3.548.
 - B. 15.045.
 - C. 101.793.
2. A bond pays a semiannual fixed coupon of 4.70%. It trades at par on its coupon date of 16 December 2025 and matures on 16 December 2033. The bond's annualized convexity statistic is *closest* to:
 - A. 51.670.
 - B. 53.231.
 - C. 206.681.
3. For changes in yield-to-maturity, the convexity adjustment is *most* needed to account for the:
 - A. first-order effect on bond prices.
 - B. bond price risk due to small changes in yield-to-maturity.
 - C. non-linear relationship of bond prices and yield to maturity.

The following information relates to questions 4-9

A bond portfolio consists of the following three option-free, fixed-rate bonds:

Bond	Market Value (£)	Annual Modified Duration	Annual Convexity
X	5,000,000	3.6239	16.2513
Y	5,000,000	9.0036	91.0278
Z	10,000,000	12.7512	179.8591

4. If the yield-to-maturity of Bond X increases by 50 bps, the expected percentage price change of Bond X is *closest* to:
 - A. 1.792%.
 - B. 1.812%.
 - C. 1.832%.
5. Given a 75 bps change in the yields-to-maturity for Bonds Y and Z, the convexity adjustment for Bond Z would be greater than the convexity adjustment of Bond

Y:

- A. if the YTM change is positive.
 - B. if the YTM change is negative.
 - C. regardless of the direction of the change in YTM.
6. For a 100 bps increase in yield-to-maturity and using money duration and money convexity, the estimated change in Bond Y's full price is *closest* to:
- A. GBP472,937.
 - B. GBP450,180.
 - C. GBP427,423.
7. For a 100 bps increase in yield-to-maturity and using the weighted average duration and convexity, the expected percentage price change for the bond portfolio is *closest* to:
- A. 7.981%.
 - B. 8.949%.
 - C. 9.533%.
8. The bond portfolio's benchmark is a fixed-income index with a duration of 9.5325 and convexity of 103.0677. Based on the weighted-average portfolio duration and convexity, the portfolio should outperform its benchmark in which of the following scenarios?
- A. Only when interest rates are rising
 - B. Only when interest rates are falling
 - C. Both when interest rates are rising and falling
9. The method of using weighted-average portfolio duration and convexity measures to assess price risk of a bond portfolio is *best* characterized as:
- A. being theoretically correct.
 - B. being commonly used by portfolio managers.
 - C. accommodating non-parallel shifts in the yield curve.
-

SOLUTIONS

1. B is correct.

$$\text{ApproxCon} = \frac{100.177546 + 99.82283 - (2 \times 100)}{(0.005)^2 \times 100} = 15.04.$$

2. A is correct.

Eight-Year, 4.70% Semiannual Bond

Fixed Coupon (%)	4.70
Periods p.a.	2
Price (per 100 Par Value)	100
Yield-to-Maturity (%)	4.70

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6 = Col. 2 × Col. 5	Col. 7 = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM/2) ⁻²
Period	Time to Receipt	Cash Flow	Present Value	Weight	Time to Receipt × Weight	Convexity of Cash Flows
1	1.0	2.35	2.296043	0.02296	0.02296	0.04384
2	2.0	2.35	2.243325	0.02243	0.04487	0.12849
3	3.0	2.35	2.191817	0.02192	0.06575	0.25108
4	4.0	2.35	2.141492	0.02141	0.08566	0.40886
5	5.0	2.35	2.092323	0.02092	0.10462	0.59920
6	6.0	2.35	2.044282	0.02044	0.12266	0.81962
7	7.0	2.35	1.997344	0.01997	0.13981	1.06774
8	8.0	2.35	1.951484	0.01951	0.15612	1.34129
9	9.0	2.35	1.906677	0.01907	0.17160	1.63811
10	10.0	2.35	1.862899	0.01863	0.18629	1.95617
11	11.0	2.35	1.820126	0.01820	0.20021	2.29351
12	12.0	2.35	1.778336	0.01778	0.21340	2.64827
13	13.0	2.35	1.737504	0.01738	0.22588	3.01871
14	14.0	2.35	1.697610	0.01698	0.23767	3.40315
15	15.0	2.35	1.658632	0.01659	0.24879	3.80002
16	16.0	102.35	70.580104	0.70580	11.29282	183.26330
			100.0000	1.00000	13.51910	206.68136
Annualized Convexity						51.67034

3. C is correct. The convexity adjustment is a complementary risk measure to duration. It accounts for the second-order (non-linear) effect of yield changes on price. It is most useful for large yield changes, because duration provides a good approximation for small yield changes.
4. A is correct. The expected price change is calculated as follows:

$$\% \Delta PV^{Full} \approx (-3.6239 \times 0.005) + [\frac{1}{2} \times 16.2513 \times (0.005)^2]$$

$$= -1.79164\% \approx -1.792\%.$$

5. C is correct. Since the convexity adjustment uses the square of the change in yield, it is always positive regardless of the direction of the change in yield-to-maturity. As a result, the convexity adjustment for Bond Z will always be greater than the convexity adjustment for Bond Y, given the same change in yields-to-maturity.

6. C is correct.

$$\text{MoneyDur} \approx 9.0036 \times 5,000,000 = 45,018,000$$

$$\text{MoneyCon} \approx 91.0278 \times 5,000,000 = 455,139,000$$

$$\Delta PV^{\text{Full}} \approx -(45,018,000 \times 0.01) + [\frac{1}{2} \times 455,139,000 \times (0.01)^2] = -\text{GBP}427,423.$$

7. B is correct. The portfolio weights for Bonds X, Y, and Z are 0.25, 0.25, and 0.50, respectively. The weighted average duration and convexity measures are calculated as follows:

$$\begin{aligned} \text{Weighted-average duration} &= (0.25 \times 3.6239) + (0.25 \times 9.0036) + (0.50 \times 12.7512) \\ &= 9.5325. \end{aligned}$$

$$\begin{aligned} \text{Weighted-average convexity} &= (0.25 \times 16.2513) + (0.25 \times 91.0278) + (0.50 \times 179.8591) \\ &= 116.7493. \end{aligned}$$

$$\% \Delta PV^{\text{Full}} \approx (-9.5325 \times 0.01) + [\frac{1}{2} \times 116.7493 \times (0.01)^2] = -8.9487\% \approx -8.949\%.$$

8. C is correct. The portfolio has a weighted-average duration of 9.5325, which is identical to the benchmark's duration. However, the portfolio has higher convexity (116.7493) compared to the benchmark (103.0677). All else equal, the portfolio should outperform the lower convexity benchmark portfolio in both rising and falling interest rate environments.
9. B is correct. The weighted-average portfolio duration and convexity method is easy to calculate and apply in practice and is commonly used by portfolio managers to assess bond portfolio price risk. It does, however, implicitly assume parallel shifts in the yield curve. Using the weighted average of time to receipt of the aggregate cash flows is the theoretically correct method to calculate portfolio duration and convexity, but it is difficult to use in practice.

PRACTICE PROBLEMS

The following information relates to questions 1-8

An investor's well-diversified portfolio has \$200,000 in cash. The investor aims to invest in short-term, one-year Large-Cap Company bonds, prior to using the cash to invest in an upcoming IPO. There are currently two Large-Cap Company bonds on the market to purchase, both with one-year maturities. One of the bonds, Bond A, is a non-callable bond, while Bond B is a callable bond. As a fixed-income analyst, you are asked to conduct an analysis.

- For the two bonds offered by Large-Cap Company, which duration calculation would be the *most* appropriate to use to measure interest rate risk?
 - Modified duration
 - Macaulay duration
 - Effective duration
- For Bond B, as the benchmark yield curve declines, the slope of the line tangent to the bond flattens as the benchmark yield declines and reaches an inflection point, after which the effective convexity becomes:
 - positive.
 - negative.
 - neither positive nor negative.
- The following information on Bonds A and B was obtained:

Bond	Effective Duration	Effective Convexity
A	7.48621	29.35972
B	7.23852	-321.75618

Compare the interest rate risk of Bond A and Bond B.

- Bond A is riskier than Bond B.
 - Bond B is riskier than Bond A.
 - Bond A and Bond B have approximately the same interest rate risk.
- For the percentage price change for Bond A, given a 200 bp increase in benchmark yield, what part of the price change is of the most concern?
 - Duration
 - Convexity
 - Not able to determine with given information

5. A colleague asks whether you also considered looking at the key rate durations when comparing the interest rate risks of Bond A and Bond B. Would research into key rate durations for Bond A and Bond B help you make a better decision about the interest rate risk of the two bonds?
 - A. Yes
 - B. No
 - C. Inconclusive
 6. The investor's portfolio is diversified, and the fixed-income component of the portfolio has bonds of various maturities, with a duration of 7.48621. What would happen to the effective duration of the fixed-income component of the investor's portfolio if Bond A is added to the portfolio?
 - A. It would increase.
 - B. It would decrease.
 - C. It would stay the same.
 7. For the two bonds under consideration, would analytical duration estimates or empirical duration estimates be *most* appropriate in conducting your analysis?
 - A. Analytical duration estimates
 - B. Empirical duration estimates
 - C. Either type of estimate would be appropriate.
 8. What impact would a "flight to safety" (i.e., government bond yields falling and credit spreads widening) have on the analytical duration estimate of Bond A?
 - A. Decreased duration
 - B. Increased duration
 - C. No impact
-

SOLUTIONS

1. C is correct. Since Bond B is callable, it has an embedded option, and effective duration is the most appropriate duration measure. Effective duration also works on bonds without embedded options and would allow the analyst to compare the interest rate risk of the two bonds.
2. B is correct. Since the bond is callable, after a certain point on the yield curve, the issuer will exercise the call and the owner will not be able to realize additional gains from declines in the benchmark yield curve.
3. B is correct. Bond B is riskier since the 100 bps upward shift in the yield curve results in a greater percentage price decrease, owing to its negative effective convexity:

$$\begin{aligned}\% \Delta PV^{Full} \text{ Bond A} &\approx (-7.48621 \times 0.0200) + [\tfrac{1}{2} \times 29.35972 \times (-0.0200)^2] \\ &= -7.33941\%.\end{aligned}$$

$$\begin{aligned}\% \Delta PV^{Full} \text{ Bond B} &\approx (-7.23853 \times 0.0200) + [\tfrac{1}{2} \times -321.75618 \times (0.0200)^2] \\ &= -8.84730\%.\end{aligned}$$
4. A is correct. Typically, the effect of duration is much larger than the effect of convexity.

$$\begin{aligned}\% \Delta PV^{Full} \text{ Bond A, Duration} &\approx -7.48621 \times 0.0200 = -14.97242\%.\end{aligned}$$

$$\begin{aligned}\% \Delta PV^{Full} \text{ Bond A, Convexity Adjustment} &\approx \tfrac{1}{2} \times 29.35972 \times (-0.0200)^2 \\ &= -0.58719\%.\end{aligned}$$
5. B is correct. Since both bonds mature in one year, key rate duration analysis would not give you any additional insight, since both bonds would undergo the same shift in the curve.
6. C is correct. Bond A has the same effective duration as the portfolio effective duration.
7. B is correct. Since both Bond A and Bond B are corporate bonds, they have credit risk, and as such, empirical duration would be the most appropriate because benchmark yields and credit spreads may not be positively correlated.
8. C is correct. There would be no impact on the analytical duration estimate. However, an empirical duration estimate would be impacted. A flight to safety would result in an increase in the bond price due to the falling benchmark yield being offset by widening credit spread. This would lead to a lower empirical duration than analytical duration.

PRACTICE PROBLEMS

1. Determine the correct answers to fill in the blanks: _____ is the risk that a bond issuer's creditworthiness deteriorates, leading investors to believe the risk of default is _____ and causing the yield spreads on the issuer's bonds to _____ and the price of its bonds to _____.
2. An uncollateralized USD200,000 note has the following characteristics:
Recovery rate 95%
Exposure at default USD150,000
If the note's expected loss in the event of default is USD2,500, its probability of default is *closest to*:
A. 15%.
B. 25%.
C. 33%.
3. Determine the correct answers to fill in the blanks: While the likelihood of default for IG borrowers is typically _____ that of an HY issuer, an investor's loss in the event of a default for secured debt is usually _____ than unsecured debt due to the secondary source of repayment.
4. A bond investor is considering the credit risk components and observed yield spreads for two IG bonds of similar maturity and liquidity:

	POD	LGD	Yield Spread
Bond 1	1.25%	75%	100 bps
Bond 2	1.1%	85%	95 bps

Which of the following conclusions about the relationship between these bonds is most correct?

- A. An investor should be indifferent between purchasing Bond 1 and Bond 2, since Bond 1's higher spread is sufficient compensation for the higher POD versus Bond 2.
 - B. An investor should prefer Bond 2 over Bond 1, since they may earn a spread that is more than sufficient for assuming the credit risk.
 - C. An investor should prefer Bond 1 over Bond 2, since it offers the highest spread relative to the expected loss.
5. Pitfalls of relying solely on credit ratings when making bond investment decisions may be best described as the following:
A. Rating agencies may change their rating in response to changing market and credit conditions for a given issuer.
B. Credit agencies may issue a positive or negative outlook for an issuer as credit risk rises or falls, but a rating change is not yet warranted.
C. Ratings may involve miscalculations or unforeseen changes not fully captured in a rating agency's forward-looking analysis.

6. An investment-grade bond with modified duration of 7 and reported convexity of 0.51 increases in price by 9.93% after a yield spread change. The value of the spread change would be *closest to*:
- A. -1.5%.
 - B. 0.15%.
 - C. 1.5%.

SOLUTIONS

1. Credit migration risk (or downgrade risk) is the risk that a bond issuer's credit-worthiness deteriorates, leading investors to believe the risk of default is higher and causing the yield spreads on the issuer's bonds to widen and the price of its bonds to fall.
2. The correct answer is C.
Here we may use Equation 1 to solve for POD:

$$EL = POD \times (EE - \text{Collateral}) \times (1 - RR).$$
Substituting in the values provided:

$$USD2,500 = POD \times (USD150,000) \times (1 - 0.95)$$

$$POD = USD2,500 / (USD150,000 \times 0.05) = 0.33 = 33\%.$$
3. While the likelihood of default for IG borrowers is typically **below** that of an HY issuer, an investor's loss in the event of a default for secured debt is usually **lower** than unsecured debt due to the secondary source of repayment.
4. C is correct. We can compare Bond 1 and Bond 2 by calculating the expected loss for each and comparing it to the annual spread:

$$EL = POD \times LGD$$

$$\text{Credit Spread} \approx POD \times LGD$$
Bond 1: $EL = 0.938\% (= 1.25\% \times 75\%)$
 $\text{Spread} - EL = 6.2 \text{ bps} (= 1.00\% - 0.938\%).$
Bond 2: $EL = 0.935\% (= 1.1\% \times 85\%)$
 $\text{Spread} - EL = 1.5 \text{ bps} (= 0.95\% - 0.935\%).$
An investor should therefore choose Bond 1 given its higher expected return versus credit risk. A is incorrect as it fails to take the LGD into account. B is incorrect as although Bond 1 earns a 1.5 bps spread above the expected annual loss, it is less than the additional 6.2 bps spread above the expected loss earned by purchasing Bond 1.
5. The correct answer is C. While answers A and B are both true, they are not a pitfall of sole reliance on credit ratings when making bond investments.
6. The correct answer is A. Equation 4 estimates bond price changes based on duration, convexity, and changes in spread:

$$\% \Delta PV^{\text{Full}} = -(\text{AnnModDur} \times \Delta \text{Spread}) + \frac{1}{2} \text{AnnConvexity} \times (\Delta \text{Spread})^2.$$
With $\text{ModDur} = 7$ and reported convexity 0.51, making the necessary convexity re-scaling and substituting in values to solve for spread results in

$$0.0993 = -(7 \times \Delta \text{Spread}) + \frac{1}{2} (51) (\Delta \text{Spread})^2$$

$$\Delta \text{Spread} = -0.0135 = -1.35\%.$$
Lower spreads make the first expression in the equation positive, along with the equation's second convexity-based term. The answer must therefore involve a decline in spreads as in answers A and B.

PRACTICE PROBLEMS

- Which of the following statements best characterizes higher creditworthiness for a non-reserve currency emerging market sovereign government?
 - A country with a government budget surplus, current account surplus with high economic growth, and a freely floating currency.
 - A country with a government budget deficit, a current account deficit with moderate economic growth and a fixed exchange rate regime
 - A country with a government budget deficit, a current account deficit, low economic growth, and a fixed exchange rate regime.
- Match the indicators below to the impact (positive or negative) on the creditworthiness of a general obligation bond issued by a state government:
 - Positive
 - Negative

A. Low unemployment	
B. Low per capita income	
C. Low per capita debt	
D. Low tax base	
E. High dependence on sales tax	
F. High post-retirement obligations	

- Based upon the following data, which country is expected to be assigned the lowest sovereign credit rating?

	Country		
		Dominican Republic	El Salvador
Economic Statistics	Costa Rica		
Long-term external debt/GDP (%)	39.6	35.7	57.0
GDP growth (%)	2.06	5.07	2.39
Currency reserves (% of GDP)	13.9	8.1	12.4

- Costa Rica
 - Dominican Republic
 - El Salvador
- Gulf Investment Corporation (GIC) is an investment company incorporated in Kuwait as a Gulf Shareholding Company. GIC is equally owned by the governments of the six member states of the Gulf Cooperation Council (GCC). GIC was formed to foster economic growth, economic diversification, and capital market development across the GCC region. The credit rating for GIC should be assessed based on its status as a:
 - sovereign agency.

- B. supranational issuer.
 - C. specialized financial intermediary or public bank.
5. Which of the following statements best characterizes external strength for a non-reserve currency sovereign country?
- A. The ability to impose and enforce strict capital controls
 - B. The establishment of a fixed exchange rate regime
 - C. The ability to generate sufficient stable foreign currency cash inflows
6. Agora is an emerging market economy with restricted capital convertibility (domestic to foreign exchange is not freely convertible). It is significantly dependent on income from the tourism sector and repatriation of income by its citizens living in foreign countries. It also has a high percentage of external debt to GDP. Which of the following is the most important factor that may result in an improvement in its sovereign creditworthiness?
- A. Diversification of its foreign currency inflows
 - B. Reducing foreign currency reserves as a percentage of GDP
 - C. Devaluation of its exchange rate versus major reserve currencies

SOLUTIONS

1. A is correct. This country exhibits fiscal stability as evidenced by its government budget surplus, high economic growth, and external strength given its current account surplus and freely floating currency.

2.

A. Low unemployment	Positive
B. Low per capita income	Negative
C. Low per capita debt	Positive
D. Low tax base	Negative
E. High dependence on sales tax	Negative
F. High post-retirement obligations	Negative

- A. Positive, as low unemployment indicates a more productive and income-generating population with potential for higher tax revenue.
 - B. Negative, as low per capita income reduces the potential for tax collection and any potential tax raise.
 - C. Positive, as low per capita debt suggests a higher debt service ratio and greater capacity for additional debt, if required.
 - D. Negative, a low tax base limits the potential for tax collection and significantly weakens the potential for state revenue to meet expenses and service its debt.
 - E. Negative, as the more diversified the state's revenue stream, the higher its credit quality. Overreliance on a single revenue stream creates the potential for volatility in case of any economic disruption.
 - F. Negative, as high post-retirement obligations indicate further state indebtedness, reducing its ability to service its debt.
3. C is correct. El Salvador is expected to have the lowest credit rating due to its high long-term external debt-to-GDP ratio. While the Dominican Republic has lower currency reserves than El Salvador, it also has a lower external debt burden and higher economic growth.
 4. B is correct. GIC is a supranational issuer, as it is sponsored and owned by multiple sovereigns. Its credit rating would be assessed as a supranational based on weighted sovereign ratings of the sponsoring countries and the nature of their implicit support in case of financial distress.
 5. C is correct. A key factor influencing whether a non-reserve currency sovereign government can meet external debt obligations is external liquidity and solvency, or the ability to generate sufficient, stable foreign currency cash inflows to meet interest and principal payments on external debt. A is incorrect, as capital controls may restrict the ability to generate FX inflows. B is incorrect, as a fixed exchange rate regime may cause instability in the case of currency devaluation.
 6. A is correct. Since Agora has high external debt to GDP and is very dependent on foreign currency inflows, it should seek to diversify its sources of foreign currency to avoid financial distress in case of disruption in tourism revenue or immigrant repatriation. B is incorrect, as maintaining or increasing foreign currency reserves is important for debt service rather than reducing FX reserves. C is incorrect. While the devaluation of Agora's exchange rate versus foreign reserve currencies may make it a more affordable tourist destination, it may also

reduce the rationale for income repatriation, leading to lower foreign inflows and a reduced ability to service external debt.

PRACTICE PROBLEMS

The following information relates to questions 1-9

The following questions are related to Mojofon Holdings:

Mojofon Holdings specializes in the design, manufacturing, and marketing of its own branded smartphones. An evolving sector, the smartphone market is dominated by few large firms who together have close to 70% of the global market. Mojofon's global market share is consistently around 3%, and it is positioned as a niche player.

An industry assessment leads to the following observations:

- Competition of the smartphone industry is intense, driven by consistent technology upgrades and new models with short product life cycles. The large investment and marketing budgets of dominant players have sustained high industry concentration.
- New entrants positioned as niche producers of “cult” models like Mojofon consistently compete for market share.
- Customers are “price-takers,” as smartphones are seen as a necessity. Pricing is usually set by the top three producers, and small firms have little pricing power. Surveys show that customers defer new phone purchases unless they have damaged their phones or there is a leap in technology.
- Core mobile phone technology, such as semiconductor chips and mobile operating licenses, are concentrated among few key suppliers that command large profit margins at the expense of phone manufacturers.
- Smartphone shipment has seen low growth in recent years due to market saturation and high penetration, limited technological advancement, and an economic slowdown. There are signs the sector is maturing. Industry concentration is increasing, with the top five firms accounting for 60–70% of global shipment and industry profit.

1. Based on her smartphone industry assessment, the credit analyst concludes that Mojofon's debt service capacity is low because:
 - A. the threat of substitutes is low.
 - B. the threat of new entrants and bargaining power of customers are low.
 - C. the industry rivalry and the bargaining power of suppliers are high.
2. Based on the overall industry assessment, the credit analyst concludes that the credit conditions for Mojofon will *most likely*:
 - A. tighten because the sector is maturing, and industry concentration is increasing.
 - B. remain unchanged because Mojofon's market share is constant despite low industry growth.

- C. expand since niche market entrants are consistently seizing market share from dominant firms.
3. Which of the following will *most likely* reduce Mojofon's near-term default risk?
- A. A new policy to increase dividends on a steady basis
- B. A three-year strategy to pursue acquisitions funded by debt
- C. Proactive management of debt maturity by issuing longer-term bonds
4. The credit analyst studies the key financial information of Mojofon and that of its peers below:

Key Credit Metrics	Mojofon		Peer Average Current Year
	Prior Year	Current Year	
EBITDA	27	20	55
Revenue	121	100	305
FCF	9	-5	40
FCF after dividends	5	-7	32
Interest expense	5	8	10
Total debt	100	100	150
Total equity	120	100	250

Debt/capital for Mojofon for the current year is *closest* to:

- A. 37.5%.
- B. 50.0%.
- C. 100.0%.
5. The outstanding bonds of Mojofon contain a covenant that requires its EBITDA/Interest coverage ratio to be above 3.0. Based on coverage ratio analysis, Mojofon *most likely*:
- A. has improved its coverage ratio.
- B. is in breach of this bond covenant.
- C. has lower credit risk than its peers.
6. Based on the financial information, Mojofon's credit risk is:
- A. below that of its peers.
- B. similar to that of its peers.
- C. above that of its peers.
7. Mojofon's capital structure consists of the following:

Type	Amount
Senior secured bonds (pledged by collateral valued at 60)	50
Senior unsecured bonds	35

Type	Amount
Subordinated bonds	15
Common equity	100

Mojofon's bonds all contain a cross-default provision. If the company is to default on one of its senior unsecured bonds:

- A. the secured bonds are likely to have full recovery.
 - B. the credit loss on senior unsecured bonds will be the same as that on subordinated bonds.
 - C. the subordinated bonds will rank below shareholders, who may dictate how recovery is to be distributed.
8. Half of Mojofon's senior unsecured bonds are issued by its major operating subsidiary, which contributes over 90% of the group's cash flow and assets, and the remainder of the group's senior unsecured bonds are issued by the holding company that relies on dividend upstreamed by its subsidiaries. There is no cross-guarantee between the holding company and the subsidiary. In an event of default, it is *most likely* that:
- A. the unsecured bonds issued by the holding company will have a higher recovery rate.
 - B. the unsecured bonds issued by the operating subsidiary will have a higher recovery rate.
 - C. the recovery rate will be the same for all senior unsecured bonds since they rank *pari passu* with each other.
9. Mojofon's S&P issuer credit rating is B+. Its S&P subordinated bond issue rating is *most likely*:
- A. BB –.
 - B. B+.
 - C. B –.

SOLUTIONS

1. The correct answer is C. Based on the industry assessment:

Competitive Forces	Intensity	Capacity to Support Debt
Threat of new entrants	Low	High
Bargaining power of suppliers	High	Low
Bargaining power of customers	Low	High
Threat of substitutes	Low	High
Industry rivalry	High	Low

Industry rivalry and bargaining power of suppliers are all high intensity, implying low capacity to support debt.

2. The correct answer is A. If the industry is maturing with low growth and the market becomes further concentrated among a few dominant firms, credit conditions for smaller smartphone producers will most likely tighten.
3. The correct answer is C. Extending debt maturity proactively will help extend Mojofon's debt maturity profile and reduce near-term default risk. B is incorrect, as debt-funded acquisitions reduce future cash flow certainty and increase credit risk. A is incorrect since increasing the dividend is a shareholder-friendly action that increases leverage and credit risk.
4. The correct answer is B. $\text{Debt/capital} = \text{Debt} / (\text{Debt} + \text{Equity})$
 $= 100 / (100 + 100) = 0.5$ or 50% for Mojofon for the current year.
5. The correct answer is B. $\text{EBITDA/Interest coverage for Mojofon for the current year} = 20/8 = 2.5$. The corresponding ratio for the prior year $= 27/5 = 5.4$. Therefore, Mojofon's coverage ratio has deteriorated, and the company has breached its coverage ratio covenant (≥ 3.0). On the other hand, $\text{EBITDA/Interest among its peers} = 55/10 = 5.5$, which is much better than that of Mojofon. Based on coverage ratio analysis, Mojofon has higher, not lower, credit risk than its peers.
6. The correct answer is C. Despite the better profit measure, EBITDA margin, versus its peers, Mojofon's cash flow measures, leverage, and coverage ratios are all considerably worse than its peers. Therefore, it has higher credit risk.
7. The correct answer is A. Senior secured bonds rank the highest in the priority of claims, and their recovery depends on the value of the asset collateral. Since the value of asset collateral (60) exceeds the amount of secured bonds outstanding (50), the secured bonds are likely to have full recovery in an event of default. Senior unsecured bonds have higher priority over subordinated bonds, which in turn have higher priority over common equity in insolvency.
8. The correct answer is B. Senior unsecured bonds issued by the holding company are said to be structurally subordinated to those issued by the major operating subsidiary. Debt at the operating subsidiary will be serviced by the cash flow and assets of the subsidiary first before any funds can be upstreamed to the holding company to service debt at that level.
 Since the operating subsidiary accounts for over 90% of the group's cash flow and assets but only half of the group's senior unsecured debt, the holding company debt can only be serviced after the operating subsidiary debt is paid in full. The unsecured bonds issued by the subsidiary will most likely have higher recovery.

9. The correct answer is C. An issuer credit rating is the rating applied to a company's senior unsecured debt. The rating agency will usually apply a notching adjustment to the lower-ranked subordinated debt, whose rating should therefore be lower than that of the senior unsecured debt.

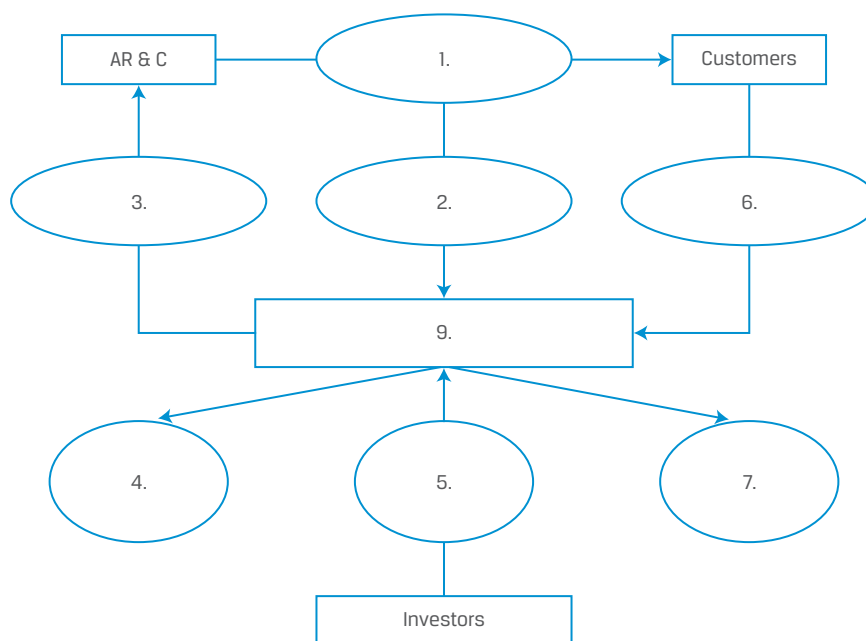
PRACTICE PROBLEMS

1. Which of the following benefits of securitization is most likely to be important for investors?
 - A. Increased efficiency
 - B. Reduced liquidity risk
 - C. The ability to tailor interest rate and credit risk exposures

The following information relates to questions 2-4

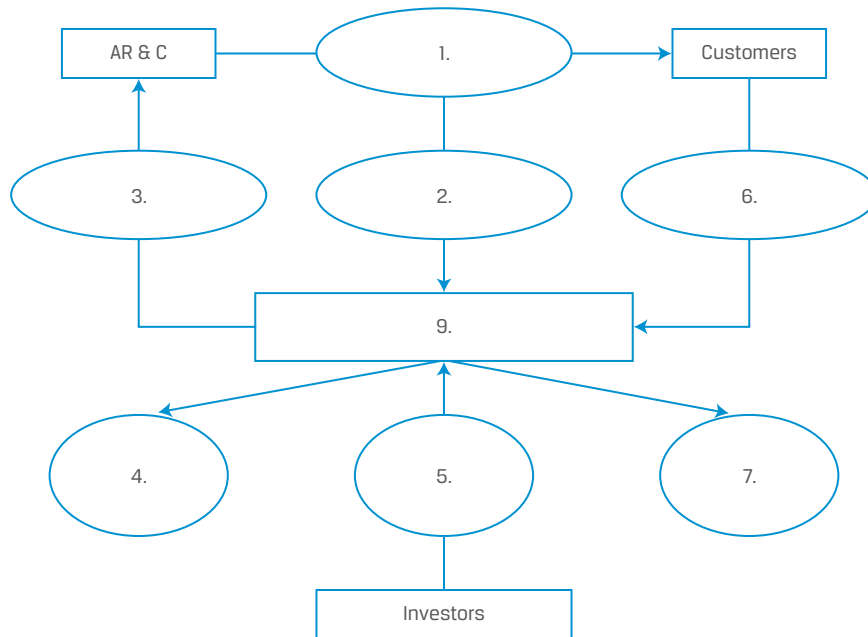
AR&C Limited leases private jets to corporations in North America for a fixed term of five years. AR&C retains ownership of the jets. The corporations make equal interest payments each month. If a customer defaults, AR&C can repossess the jet. AR&C has been in business for approximately fifteen years and has a B3 credit rating, meaning that its debt is non-investment grade. AR&C has CAD500 million in leases on its balance sheet, which it wants to securitize into ABS.

2. Determine which of the following is the *best* next step for AR&C.
 - A. AR&C issues and sells securities backed by the pool of leases.
 - B. AR&C sets up a separate legal entity to which it sells the CAD500 million in leases.
 - C. AR&C discontinues servicing the leases, shifting this responsibility to a special purpose company.
3. Which shape *best* represents the cash payments for the Leases?



- A. Shape 2
- B. Shape 3
- C. Shape 5

4. Which of the following *best* describes the entity shown in shape 9?



The entity in shape 9 is the:

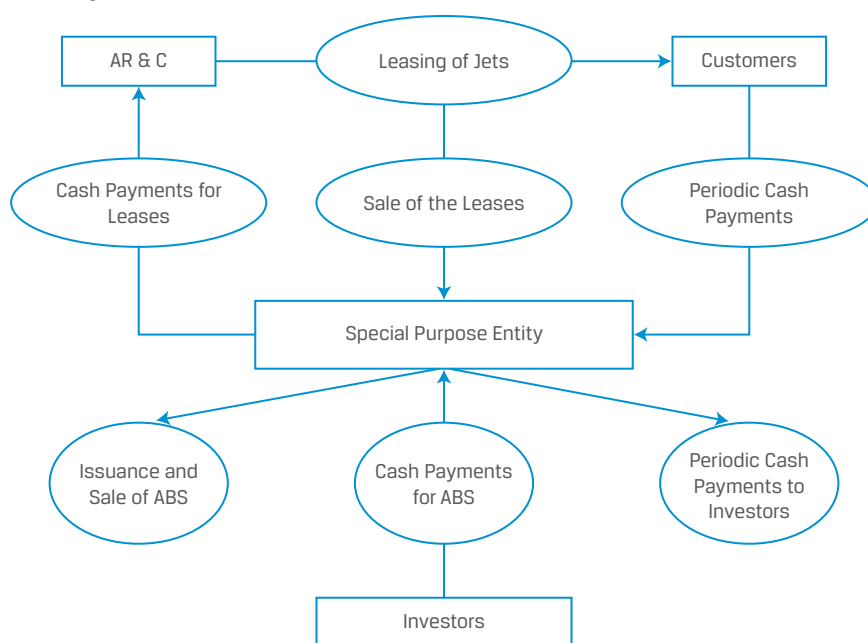
- A. seller of the collateral.
 - B. servicer of the leases.
 - C. SPE that purchases the leases and uses them as collateral for the ABS.
-
5. Which of the following statements about securitization's impact on economies and financial markets is correct?
- A. Securitization decreases overall liquidity in the financial system.
 - B. Securitization provides an alternative means of funding business operations beyond traditional financing.
 - C. Securitization allows companies to directly access the financial markets and purchase loans that will be held on their balance sheet.
6. Which of the following statements about securitization is incorrect?
- A. The ABS that are created by securitization have characteristics similar to those of equity investments.
 - B. Buying securitized debt helps investors increase exposure to the risk–return characteristics of a wider range of underlying assets.
 - C. Securitization transactions allow issuers to create risk–return characteristics to satisfy the risk tolerance of different investors.

7. In the event of a bankruptcy of the originator in a loan securitization, the investors would *most likely* experience:
 - A. gains.
 - B. losses.
 - C. no impact.

8. In a securitization, the party that purchases the loans or receivables and uses them as collateral to issue ABS is the:
 - A. SPE.
 - B. seller.
 - C. servicer.

SOLUTIONS

1. The correct answer is C. Securitization provides a direct conduit between borrowers and investors. It also allows investors to tailor interest rate and credit risk exposures to suit their specific risk, return, and maturity needs. A is incorrect because securitization creates efficiency for issuers and financial markets more directly than for investors. Securitization allows issuers to sell illiquid assets and operate more efficiently on a risk-adjusted basis. Moreover, because securitization creates tradable securities with higher liquidity than the original underlying assets, it makes financial markets more efficient. B is incorrect because securitization improves overall liquidity in the financial system and reduces liquidity risk, which benefits participants in financial markets.
2. The correct answer is B. In order for these assets (leases) to be securitized, they must be held in a separate legal entity structure distinct from AR&C. The separate legal entity structure ensures that if AR&C, the originator of the leases, files for bankruptcy, the leases backing the ABS that are issued by the separate legal entity are secure there, thus AR&C's creditors cannot have any claim on them. A is incorrect because if AR&C issues the securities, the leases remain on AR&C's balance sheet and thus could still be accessible to AR&C's creditors in the event of a bankruptcy. C is incorrect because servicing the leases must continue whether or not by AR&C or another entity, but servicing the leases is separate from securitizing them.
3. The correct answer is B. As shown in the diagram below, the SPE (in rectangle 9 above), pays AR&C for the leases (as noted in shape 3 above). A is incorrect because shape 2 above represents the sale of the leases (as shown in the diagram below), not the payment for them, and C is incorrect because shape 5 above represents payments from the investors for the ABS sold by the SPE (as shown in the diagram below).



4. The correct answer is C. The SPE purchases the leases from AR&C and uses them as collateral for the ABS, which it issues and sells to the investors. A is incorrect because AR&C is the seller of the collateral, not the SPE. C is incorrect because

the diagram does not show clearly who services the leases, but it does show that the entity in shape 9 is the SPE.

5. The correct answer is B. An important benefit of securitization is that it provides an alternative means of funding business operations beyond traditional financing tools, such as bonds, preferred equity, and common equity. A is incorrect because securitization improves overall liquidity in the financial system and reduces liquidity risk. C is incorrect because securitization allows companies to directly access the financial markets and move loans off their balance sheet.
6. The correct answer is A. The ABS that are created by pooling loans and receivables have characteristics similar to those of a standard bond. B is incorrect because it is a correct statement. Holding securitized debt helps pension fund investors increase exposure to the risk–return characteristics of a wider range of underlying assets. C is incorrect because it is a correct statement. Securitization transactions allow issuers to design risk–return characteristics to satisfy the risk tolerance of different investors.
7. The correct answer is C. Since the SPE, which owns the loans, is bankruptcy remote from the seller of the collateral, the bankruptcy of the seller of the collateral will not affect the holders of securities issued by the SPE and backed by the collateral. The only credit risk that the investors face is the risk that the borrowers whose loans are included in the SPE default on their loans.
8. The correct answer is A. The SPE is often referred to as the issuer in the prospectus because it is the entity that issues the securities to investors. B is incorrect because the seller of the collateral, also called the depositor, is the party that originally aggregated the assets and sold them to the SPE. C is incorrect because the servicer has an administrative function and is the entity that collects payments from borrowers.

PRACTICE PROBLEMS

1. Which of the following statements regarding securitized products is correct?
 - A. Credit tranching offers credit protection for the equity tranche in a securitization.
 - B. Pass-through securities are the simplest securitized structure and involve issuers retaining the underlying assets.
 - C. In a true securitization, the underlying pool of assets is removed from the balance sheet and transferred into an independent legal entity that issues securities backed by these assets.
2. Upon a bankruptcy affecting a covered bond, the first available safeguards to protect against potential losses are the:
 - A. ringfenced loans.
 - B. unencumbered assets of the issuer.
 - C. assets added by the collateral manager during ramp-up.
3. Consider the following potential reasons to explain why covered bonds usually carry lower credit risks and offer lower yields than otherwise similar ABS:
 1. Eligibility criteria
 2. Dynamic cover pool
 3. Redemption regime in event of sponsor defaultWhich of the following factors accurately explain(s) this result?
 - A. Reason 1 only
 - B. Reasons 1 and 2 only
 - C. Reasons 1, 2, and 3
4. Select the internal credit enhancement that seeks to improve the overall credit quality of a risky pool of securitized loans.
 - A. Letters of credit
 - B. Overcollateralization
 - C. Cash collateral accounts
5. For an issuer, which of the following outcomes is *most likely* after offering a credit card ABS?
 - A. Increased cost of funding
 - B. Increased income from fees
 - C. Increased cost of default risk
6. An action affecting the cash flow received by a credit card ABS holder during its

revolving period is the:

- A. early repayment of principal by cardholders.
 - B. card's floating-rate cap exceeding the periodic rate.
 - C. triggering of an ABS rapid amortization provision.
7. A characteristic of solar loans that makes them attractive to potential solar ABS investors is their:
- A. universal availability to any homeowners.
 - B. ability to combine multiple liens to mitigate default risk.
 - C. flexibility in allowing either purchase or rental of solar systems.
8. Which investor tranche plays a key role in determining CLO viability?
- A. Senior
 - B. Equity
 - C. Mezzanine
9. The inclusion of assets into a CLO collateral pool is completed:
- A. during a subsequent ramp-up period.
 - B. prior to the close of the CLO transaction.
 - C. between ramp-up and loan maturity on meeting certain requirements.
10. When a CLO transaction experiences a collateral pool default:
- A. senior tranche holders are guaranteed full repayment.
 - B. call features embedded in bonds likely affect junior tranches.
 - C. losses are distributed proportionately across all the tranches.
11. The feature of a covered bond transaction *most likely* shared with both CDOs and non-mortgage ABS is its:
- A. specified LTV cutoff.
 - B. multiple tranches for the cover pool.
 - C. distinct maturity and settlement dates.
12. Which of the following statements about CDOs is correct?
- A. The collateral pools for CDOs are static.
 - B. The proceeds to pay the CDO bond classes can only come from interest payments from collateral assets.
 - C. A CDO is a leveraged transaction, where equity tranche holders use borrowed funds to generate a return above the funding cost.

SOLUTIONS

1. The correct answer is C. In a true securitization, the specific pool of assets is removed from the balance sheet and transferred into a separate and independent legal entity that then issues securities backed by these pooled assets. A is incorrect because credit tranching offers credit protection for the more senior bond classes in a securitization not the equity tranche. B is incorrect because covered bonds, not pass-through securities, are the simplest securitization structure, which involves the issuer retaining the underlying assets on its balance sheet.
2. The correct answer is A. In the case of bankruptcy covered bond investors, they have dual recourse with the first safeguard being the ringfenced loans in the cover pool that underlie the covered bond transaction. B is incorrect because while investors also have recourse to the unencumbered assets of the issuer, those serve as a subsequent safeguard. C is incorrect because this post-transaction contribution is characteristic of the non-amortizing structure of a CDO (collateralized debt obligation).
3. The correct answer is C. All three of the listed reasons explain this comparative result, which has made covered bonds a relatively stable and reliable source of funding over time.
4. The correct answer is B. Overcollateralization, one of the three main types of internal credit enhancements used in securitization transactions, is a provision that the collateral underlying the transaction is larger than the face value of the issued bonds. The extra collateral provides a cushion to cover losses from defaults in the underlying pool of collateral. A and C are incorrect because both are types of external, not internal, credit enhancements.
5. The correct answer is B. A credit card ABS generates additional fee income. A is incorrect because by removing the credit card debt from the balance sheet, ABS issuance improves capital efficiency and lowers the cost of funding. C is incorrect as the ABS also reduces the cost of default risk from credit card debt.
6. The correct answer is C. Triggering rapid amortization provisions accelerates and alters principal cash flows. Noteholders will receive their investments earlier, which they can then reallocate to other investments offering more attractive risk/return characteristics. A is incorrect because typically during the revolving period, as credit cards are non-amortizing loans, the security holders receive only payments from the finance charges and fees the lender collects. Returned principal is reinvested in the collateral pool. B is incorrect because when floating-rate cap exceeds the rate cardholders must pay, there is no related binding limit on finance payments.
7. The correct answer is B. Default risk minimization adds to investment appeal, and solar loans provide opportunities for overcollateralization. Normally, solar ABS are collateralized by the underlying debt: mortgages, loans, or receivables. These loans can be further collateralized by a lien pledged on the installed systems, on the property itself, or both. A is incorrect as solar loans are typically extended only to prime borrowers who own their homes and have good payment records. C is incorrect as the loans enable property owner purchase of installed systems instead of rental that is transacted through solar leases.
8. The correct answer is B. Investors in equity tranches take on equity-like risks with the potential to earn returns comparable to equities. Moreover, these

residual tranche investors play a key role in whether a CLO is viable or not; the CLO structure has to offer competitive returns for this tranche. Facing these risk/return possibilities puts them in the position of the marginal, price-setting investors. A and C are incorrect because they are promised fixed returns without reference to the underlying performance of the collateral pool. They are also exposed to less risk in having seniority payoff preference in case of default.

9. The correct answer is C. After the ramp-up period but before underlying collateral pool loans mature, the collateral manager may replace loans in the portfolio as long as the new asset meets the portfolio selection criteria. A is incorrect because while additional assets are added to the collateral pool during a subsequent ramp-up period, loan replacements can still occur after the ramp-up period. B is incorrect because a typical feature of CLO transactions is that the collateral portfolio is not finalized until after the transaction closes.
10. The correct answer is B. A complicating factor of a default can be the call features embedded in bonds. While exercising call options on debt would replenish the collateral pool, distributing these proceeds to the senior and mezzanine tranches would shrink the size of the collateral pool, further narrowing the earnings potential with junior tranches receiving the least cash flow. A is incorrect because senior tranches are then in danger of not receiving some or any payments, depending on the size of the default. C is incorrect because losses would first be absorbed by junior tranche investors, with remaining losses realized by the other tranches in order of repayment seniority.
11. The correct answer is C. As illustrated by their respective term sheets, all three types of ABS transactions have timing milestones designating due dates for investments (settlement date) and the end of their terms (maturity date). A is incorrect because in contrast to their non-amortizing counterparts, covered bonds are backed by a segregated pool of assets that typically consist of commercial or residential mortgages, for which an LTV cutoff is applicable. B is incorrect because while other ABS often use credit tranching to create bond classes with different borrower default exposures, covered bonds usually consist of one bond class per cover pool.
12. The correct answer is C. A CDO is a leveraged transaction, where equity tranche holders use borrowed funds (i.e., the bond classes issued) to generate a return above the funding cost. A is incorrect because unlike MBS, the pools in a CDO are not static; so, there is a need for a collateral manager that buys and sells debt obligations for and from the CDO's collateral pool to generate sufficient cash flows to meet the obligations to the CDO bondholders. B is incorrect because the proceeds to pay the CDO bond classes can come from interest payments from collateral assets, maturing of collateral assets, and sale of collateral assets.

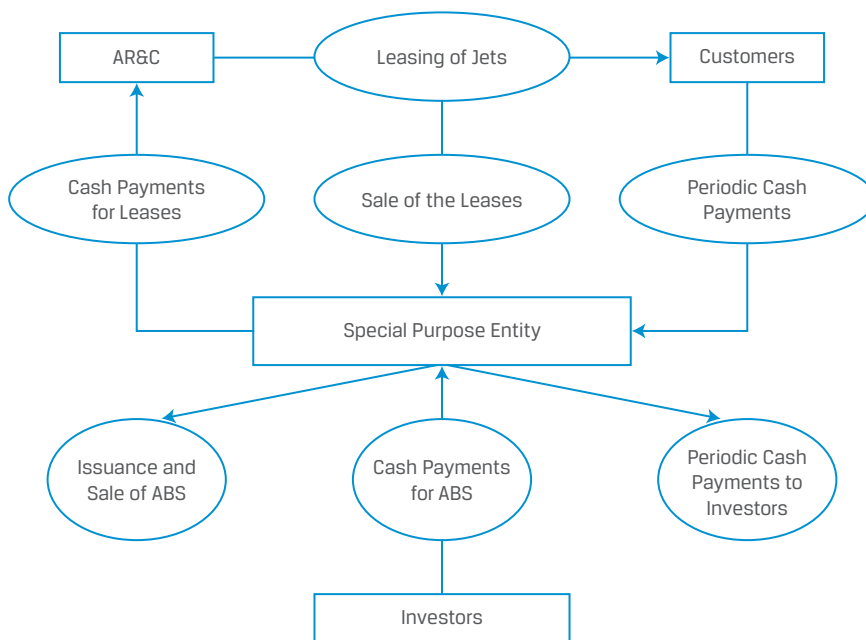
PRACTICE PROBLEMS

The following information relates to questions 1-2

Please refer to the information below to answer questions 1 and 2 regarding the bonds securitized by the AR&C leases.

AR&C Limited Securitization of Private Jet Leases

AR&C Limited leases private jets to corporations in North America for a fixed term of five years. AR&C retains ownership of the jets. The corporations make equal interest payments each month. If a customer defaults, AR&C can repossess the jet. AR&C has been in business for approximately fifteen years and has a B3 credit rating, meaning that its debt is non-investment grade. AR&C has CAD500 million in leases on its balance sheet that it wants to securitize into ABS.



Bond Class

A (senior)	<i>Tranche A Notes</i> Face value: CAD250 million Interest rate: MRR + 1.00% Maturity: One year
B (subordinated)	<i>Tranche B Notes</i> Face value: CAD100 million Interest rate: MRR + 2.00% Maturity: Two years
C (subordinated)	<i>Tranche C Notes</i> Face value: CAD100 million Interest rate: MRR + 3.00% Maturity: Three years

Bond Class	
D (subordinated)	<i>Tranche D Notes</i> Face value: CAD50 million Interest rate: MRR + 4.00% Maturity: Four years
Total	CAD500 million

- Compare the credit rating of the Class A bonds issued by the SPE and the un-collateralized bonds issued directly by AR&C. The Class A bonds are *most likely* rated:
 - lower than AR&C bonds.
 - the same as AR&C bonds.
 - higher than AR&C bonds.
- Select which of the following statements related to the collateralized bonds issued directly by the SPE is *most* accurate.
 - This senior/subordinated structure is an example of time tranching.
 - Losses are realized by the subordinated bond classes before any losses are realized by Class A bonds.
 - In a waterfall structure such as this one, losses are shared proportionally across all subordinated bond classes.

The following information relates to questions 3-4

A borrower wishes to procure a mortgage loan to purchase a property. Information regarding the property, the desired loan, and the buyer's finances is shown below:

- Property price: CAD750,000
- Desired loan amount: CAD525,000
- Annual interest rate: 5.00%
- Loan term: 30 years/360 months
- Monthly payment: CAD2,818.31
- Borrower's annual pre-tax gross income: CAD210,000

- The loan-to-value (LTV) ratio is *closest to*:
 - 54%.
 - 66%.
 - 70%.

4. The debt-to-income (DTI) ratio is *closest to*:
- A. 1.30%.
 - B. 16.10%.
 - C. 250.00%.
-
5. For a CMO that includes Planned Amortization Class (PAC) tranches, if the prepayment rate is within the anticipated range, which of the following tranches *most likely* protects investors from prepayment risk?
- A. PAC tranche
 - B. Senior tranche
 - C. Support tranche
6. Which of the following approaches redistributes prepayment risk?
- A. Tranching
 - B. Excess spread
 - C. Overcollateralization
7. Shown below are five mortgages that comprise the entire pool in a mortgage-backed security.

Mortgage	Interest Rate (i)	Beginning Balance (BB)	Current Balance (CB)	Original Term (months)	Number of Months to Maturity
A	2.00%	CAD200,000	CAD83,000	420	140
B	3.00%	CAD300,000	CAD193,000	360	192
C	4.00%	CAD400,000	CAD315,000	360	240
D	5.00%	CAD500,000	CAD357,000	240	144
E	6.00%	<u>CAD600,000</u>	<u>CAD548,000</u>	420	324
		CAD2,000,000	CAD1,496,000		

The WAM for this security is *closest to*:

- A. 208 months.
 - B. 224 months.
 - C. 236 months.
8. Which of the following adverse consequences is most likely associated with extension risk?
- A. Investors must reinvest the proceeds at lower interest rates.
 - B. Higher interest rates reduce the value of the cash flows investors receive.
 - C. The prepayment option reduces the potential price appreciation for the bond.

The following information relates to questions 9-11

Use the following information for questions 9, 10, and 11:

Shown below are three properties whose mortgages are securitized to create a sequential-pay CMBS and information related to the four tranches comprising the CMBS.

Property Information					
Property Type	Fair Market Value	Mortgaged Amount	Coupon Rate	Maturity	Repayment
Office Building	35,000,000	27,000,000	4.00%	Four years from settlement date	Fully amortizing
Office Building	60,000,000	44,000,000	3.00%	Four years from settlement date	Fully amortizing
Warehouse	15,000,000	12,000,000	5.00%	Eight years from settlement date	Fully amortizing

Prepayment: No prepayment on the underlying mortgages is allowed in the first 5 years.

Tranche Information		
Tranche	Coupon Rate	Principal Value
A	2.00%	20,000,000
B	3.00%	22,000,000
C	4.00%	23,000,000
D	5.00%	18,000,000

9. The associated loan to value ratio is *closest to*:
 - A. 0.75.
 - B. 0.77.
 - C. 0.80.
10. The weighted average coupon payment is *closest to*:
 - A. 3.47%.
 - B. 3.50%.
 - C. 3.61%.
11. The weighted average proceeds from the mortgages is *closest to*:
 - A. 3.47%.
 - B. 3.50%.

C. 3.61%.

12. Which of the following would *most likely* increase balloon risk for a commercial mortgage loan that is maturing?
- A. High DSCR
 - B. Low LTV ratio
 - C. Lack of property buyers in market
13. Which of the following provides call protection for CMBS investors at the structural level?
- A. Defeasance
 - B. Prepayment lockout
 - C. Sequential-pay tranches

SOLUTIONS

1. The correct answer is C. The Class A bonds issued by the SPE are backed by collateral, the leases sold by AR&C to the SPE, and they are the most senior of the bonds issued by the SPE. A secured bond is secured by collateral and is considered less risky than a bond without collateral backing. The Class A bonds issued by the SPE would *most likely* be rated higher than any unsecured corporate bonds issued directly by AR&C.
2. The correct answer is B. Losses are realized by the subordinated bond classes before any losses are realized by the senior Class A bonds. A is incorrect because this senior/subordinated structure is an example of credit tranching, not time tranching. C is incorrect because in a waterfall structure such as this one, the most junior tranche, Class D, will absorb all losses up to its full CAD50 million par value first. Then, if credit losses exceed that threshold, Bond Class C will absorb all losses up to its full CAD100 million par value. The losses are not shared proportionally across all subordinated bond classes; instead, they are absorbed by the most junior classes first and subsequently by the more senior ones up to the value of the loss.
3. The correct answer is C. The loan-to-value ratio (LTV) is the ratio of the amount of the loan/mortgage to the property's value.

$$LTV = \frac{\text{Amount of the loan/mortgage}}{\text{Property's value}}$$

$$LTV = \frac{525,000}{750,000}$$

$$LTV = 70\%.$$

A is incorrect because it was calculated as the monthly payment divided by the beginning loan value, not the loan to property value, as follows:

$$\frac{2,818.31}{525,000} = 0.54\%.$$

Note: This is 0.54%, not 54%. Choice A was adjusted to 54% to make the correct answer less obvious.

B is incorrect because it was calculated as the monthly payment for a 20-year loan, not a 30-year loan, divided by the beginning loan value, not the loan to property value, as follows:

$$\frac{3,464.77}{525,000} = 0.66\%.$$

Note: This is 0.66%, not 66%. Choice B was adjusted to 66% to make the correct answer less obvious.

4. The correct answer is B. The DTI is calculated using the monthly debt payment and the borrower's monthly pre-tax gross income, as follows:

$$DTI = \frac{\text{Monthly Debt Payment}}{\text{Monthly pre-tax gross income}}$$

$$DTI = \frac{\text{CAD } 2,818.31}{(\text{CAD } 210,000/12)} = 16.10\%.$$

A is incorrect because it was calculated as the monthly payment divided by the annual pre-tax gross income, as follows:

$$\frac{2,818.31}{210,000} = 1.30\%.$$

C is incorrect because it was calculated as the starting loan value divided by the annual pre-tax gross income, as follows:

$$\frac{525,000}{210,000} = 250.00\%.$$

5. The correct answer is C. If the prepayment rate is within the specified range, all prepayment risk is absorbed by the support tranche. This provides greater predictability of the size and timing of cash flows paid to investors in the PAC tranches.
A is incorrect because in a CMO that includes a PAC tranche, the PAC tranche is protected from prepayment risk because all prepayment risk is absorbed by the support tranche.
B is incorrect because in a CMO that includes a PAC tranche, the senior (PAC) tranche is protected from prepayment risk because all prepayment risk is absorbed by the support tranche.
6. The correct answer is A. The tranching structure in CMOs cannot eliminate prepayment risk, but it can redistribute prepayment risk across the different tranches, insulating some tranches more than others. CMO tranche structures reduce the uncertainties of the size and timing of payments investors receive.
B and C are incorrect because excess spread and overcollateralization are both common forms of credit enhancement, but they do not redistribute prepayment risk.
7. The correct answer is C. The weighted average maturity is calculated by weighting the number of months to maturity for each mortgage in the pool by the percentage of the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool, as shown in the following equation:

$$\begin{aligned} WAM &= MM_A \left(\frac{CB_A}{\sum CB} \right) + MM_B \left(\frac{CB_B}{\sum CB} \right) + MM_C \left(\frac{CB_C}{\sum CB} \right) \\ &+ MM_D \left(\frac{CB_D}{\sum CB} \right) + MM_E \left(\frac{CB_E}{\sum CB} \right) \\ WAM &= 140 \left(\frac{83,000}{1,496,000} \right) + 192 \left(\frac{193,000}{1,496,000} \right) + 240 \left(\frac{315,000}{1,496,000} \right) \\ &+ 144 \left(\frac{357,000}{1,496,000} \right) + 324 \left(\frac{548,000}{1,496,000} \right) \end{aligned}$$

$$WAM = 236 \text{ months.}$$

A is incorrect because it is calculated by averaging the months to maturity for the five mortgages, rather than calculating their weighted average.

B is incorrect because it is calculated by weighting the number of months to maturity for each mortgage in the pool by the percentage of the beginning mortgage balance relative to the beginning amount of all the mortgages in the pool, not by the current balances.

8. The correct answer is B. Extension risk is the risk that the borrower might pay back the money borrowed more slowly than anticipated, extending the time of repayment and the maturity of the bond. For investors, this has one adverse consequence: Higher interest rates reduce the value of the cash flows investors receive. The payments the investors receive will be discounted at a higher interest

rate, and the extension stretches out the payments the investors receive.

A is incorrect because contraction risk (the risk that the borrower might pay back the money borrowed more quickly than anticipated, reducing the amount of future payments the investor receives), not extension risk, has the adverse consequence that investors must reinvest the proceeds at lower interest rates.

C is incorrect because contraction risk (the risk that the borrower might pay back the money borrowed more quickly than anticipated, reducing the amount of future payments the investor receives), not extension risk, has the adverse consequence that the prepayment option reduces the potential price appreciation for the bond.

9. The correct answer is A. The associated loan to value ratio is calculated as the total value of all loans (mortgages) divided by the total value of the properties associated with the mortgages, as follows:

$$LTV = \frac{\text{Total value of all loans (mortgages)}}{\text{Total value of all properties mortgaged}}$$

$$LTV = \frac{83,000,000}{110,000,000} = 0.75.$$

B is incorrect because 0.77 is the value of just the mortgage on the first property (27,000,000) divided by the first property's total value (35,000,000).

C is incorrect because 0.80 is the value of just the mortgage on the third property (11,400,000) divided by the third property's total value (15,000,000).

10. The correct answer is A. The weighted average coupon payment is calculated as the value of each tranche's coupon rate weighted by the principal value of that tranche as a percentage of the total value of all tranches:

$$WAC = i_A \left(\frac{PV_A}{\sum PV} \right) + i_B \left(\frac{PV_B}{\sum PV} \right) + i_C \left(\frac{PV_C}{\sum PV} \right) + i_D \left(\frac{PV_D}{\sum PV} \right)$$

$$WAC = 2.00\% \left(\frac{20,000,000}{83,000,000} \right) + 3.00\% \left(\frac{22,000,000}{83,000,000} \right) + 4.00\% \left(\frac{23,000,000}{83,000,000} \right) + 5.00\% \left(\frac{18,000,000}{83,000,000} \right)$$

$$WAC = 3.47\%.$$

B is incorrect because 3.50% is calculated as a straight average of the four coupon rates (of the four tranches), rather than a weighted average.

C is incorrect because 3.61% is the weighted proceeds from the three mortgages, not the weighted average coupon payment.

11. The correct answer is C. The weighted average proceeds from the mortgages (WAMP shown below) is calculated as the coupon rate of each mortgage weighted by that mortgage's value/amount as a percentage of the total value of all the mortgages:

$$WAMP = i_A \left(\frac{MA_A}{\sum MA} \right) + i_B \left(\frac{MA_B}{\sum MA} \right) + i_C \left(\frac{MA_C}{\sum MA} \right)$$

$$WAMP = 4.00\% \left(\frac{27,000,000}{83,000,000} \right) + 3.00\% \left(\frac{44,000,000}{83,000,000} \right) + 5.00\% \left(\frac{12,000,000}{83,000,000} \right)$$

$$WAMP = 3.61\%.$$

A is incorrect because 3.47% is the weighted average coupon payment for the four tranches, not the weighted average proceeds from the mortgages.

B is incorrect because 3.50% is calculated as a straight average of the four coupon rates (of the four tranches), not the weighted average proceeds from the mortgages.

12. The correct answer is C. A lack of property buyers in the market will make it challenging for a borrower to sell a property to generate sufficient funds to pay off the outstanding principal balance. A is incorrect because a high DSCR would reduce the balloon risk. B is incorrect because a low LTV ratio would also reduce the balloon risk of a maturing loan.
13. The correct answer is C. Structural call protection is achieved through sequential-pay tranches in the CMBS as a lower-rated tranche cannot be paid down until the higher-rated tranche is completely retired. A is incorrect because defeasance provides call protection at the individual loan level. Defeasance allows prepayment, but the borrower must purchase a portfolio of government securities that fully replicates the cash flows of the remaining scheduled principal and interest payments, including the balloon loan balance, on the loan. B is incorrect because prepayment lockout provides call protection at the individual loan level. Prepayment lockout is a contractual agreement that prohibits any prepayments during a specified period.

